

Mathematical background

Branching, quasi-stationarity and absorption models

Adam Aldridge

Chapter 2
University of Leeds

Contents

1	Introduction	3
2	The Galton–Watson process	3
2.1	Death or division	4
2.2	Certainty of extinction	6
2.3	The critical case: a power law and an infinite expected lifetime	6
3	The limiting conditional distribution	8
3.1	Late-time survival probability	8
3.2	Mean	9
3.3	Variance	9
3.4	A continuum view of the survival recursion	12
4	Survival of small populations	12
4.1	An initial cohort of size k	13
4.2	An aside on early replication	13
4.3	Discrete birth–death–catastrophe	15
4.4	Continuous-time birth–death	16
5	The constant $A(p)$	17
5.1	An infinite-product representation	17
5.2	An exact series and two-sided bounds	19
5.3	Behaviour as $p \rightarrow \frac{1}{2}^-$	20
5.4	Why the discrete and continuous constants differ	22
5.5	Searching for a closed form	22
5.6	The Koenigs connection	24
5.7	Computing $A(p)$ in practice	27
6	Absorption models and the method of characteristics	29
6.1	Absorption only	29

6.2	Absorption–death	32
6.3	The method of characteristics, tested on absorption–death	33
6.4	Absorption–birth–death	34
Appendices		37
A	An integral identity for the hypergeometric function	37
B	Extracting state probabilities from a generating function	39

1 Introduction

The models used throughout this thesis describe populations that are small enough for chance to matter. A handful of bacteria deposited in a mammalian host, a few viral particles inside a single cell: in each case the population is far from the regime where deterministic rate equations are informative, and the questions that matter — will the population establish itself at all? if it survives, how large will it be? — are questions about the fluctuations rather than about the mean. This chapter assembles the mathematical apparatus needed to answer them.

Two threads run through what follows, and it is worth saying at the outset how they relate.

The first concerns *conditioning on survival*. A subcritical branching process dies out with probability one, so its unconditional mean decays geometrically and tells us very little. But if we look only at those realisations that happen still to be alive at generation n , a quite different picture emerges: the conditional population settles down to a fixed distribution, the *quasi-stationary* distribution, whose mean is a constant. For the binary Galton–Watson process with replication probability $p < \frac{1}{2}$ that constant is $1/A(p)$, where

$$A(p) = \lim_{n \rightarrow \infty} \frac{S_n}{(2p)^n},$$

S_n being the probability of survival to generation n . Much of this chapter is concerned with $A(p)$: what it is, how to compute it, and why — despite a good deal of effort — no elementary formula for it has been found.

The second thread concerns *the near-critical regime*. As p approaches $\frac{1}{2}$ from below the process takes ever longer to die, the quasi-stationary mean $1/A(p)$ diverges, and every numerical scheme that relies on iterating to stationarity becomes impractical. Just above $\frac{1}{2}$ the initial count of replicative individuals comes to weigh heavily on the chance of the process surviving at all. The behaviour on either side of $p = \frac{1}{2}$ is what makes small inocula interesting, and it is also what makes them awkward to compute with.

[Section 2](#) sets out the Galton–Watson process and establishes the two facts on which everything else rests: that extinction is certain at and below criticality, and that at criticality the expected lifetime is nevertheless infinite. [Section 3](#) derives the limiting conditional mean and variance and so introduces $A(p)$. [Section 4](#) looks at what changes when the process starts from a cohort of k individuals rather than one, and works out the corresponding constant for the continuous-time birth–death process, where a closed form *is* available. [Section 5](#) is devoted to $A(p)$ itself. It gives an infinite-product representation, an exact series with two-sided bounds that pin the constant between the continuous-time answer and twice it, the near-critical asymptotic $A(p) \sim 2(1 - 2p)$, and an account of why the exact function resists closed-form description — a question that turns out to be one about the Koenigs linearisation of the logistic map. [Section 6](#) changes tack and develops the machinery of generating-function partial differential equations and the method of characteristics for the absorption models used later in the thesis.

Throughout, results that are standard are presented as such, and the places where the argument is heuristic or the numerics are doing the real work are marked explicitly. Where a claim can be checked numerically, it has been.

2 The Galton–Watson process

The Galton–Watson process is a discrete-time, discrete-state Markov chain in which every member of a population independently either replicates or dies at each generational increment. Formulated by

Francis Galton and Henry Watson in the late nineteenth century, it remains the most fundamental model of stochastic branching [1, 2].

Suppose each individual produces, at the end of its life and independently of every other, a number of offspring distributed as the random variable $\mathbf{X}$, with

$$\Pr\{\mathbf{X} = k\} = p_k, \quad k = 0, 1, 2, \dots,$$

and denote the mean number of offspring per individual by $\mu = \mathbb{E}(\mathbf{X})$. Let $\mathbf{Z}_n$ be the number of individuals in the n th generation, with a single progenitor, $\mathbf{Z}_0 = 1$. Conditioning on the previous generation and using the tower property,

$$\mathbb{E}(\mathbf{Z}_{n+1}) = \mathbb{E}(\mathbb{E}(\mathbf{Z}_{n+1} \mid \mathbf{Z}_n)) = \mathbb{E}(\mu \mathbf{Z}_n) = \mu \mathbb{E}(\mathbf{Z}_n),$$

so that $\mathbb{E}(\mathbf{Z}_n) = \mu^n$. If $\mu < 1$ the expected population decays geometrically and $\lim_{n \rightarrow \infty} \Pr\{\mathbf{Z}_n = 0\} = 1$: extinction is certain.

Writing $\phi(z) = \mathbb{E}(z^{\mathbf{X}})$ for the probability generating function of the offspring distribution and $u_n = \Pr\{\mathbf{Z}_n = 0\}$ for the probability of extinction by generation n , a first-step argument gives the iteration on which almost everything in this chapter depends,

$$u_{n+1} = \phi(u_n). \tag{1}$$

2.1 Death or division

Throughout we specialise to the binary — or *simple* — case, in which an individual either divides into two or dies:

$$\begin{aligned} p_2 &= p && \text{(division),} \\ p_0 &= 1 - p && \text{(death),} \\ p_i &= 0 && \forall i \neq 0, 2. \end{aligned}$$

The mean number of offspring is then

$$\mu = 2p,$$

and the expected population descended from a single ancestor is

$$\mathbb{E}(\mathbf{Z}_n) = (2p)^n. \tag{2}$$

The process is *subcritical* when $2p < 1$, *critical* when $p = \frac{1}{2}$, and *supercritical* when $2p > 1$. [Figure 1](#) shows realisations either side of the critical point.

The offspring probability generating function is

$$\phi(z) = \mathbb{E}(z^{\mathbf{Z}_1}) = pz^2 + (1 - p), \tag{3}$$

so that (1) reads

$$u_{n+1} = pu_n^2 + 1 - p. \tag{4}$$

The recursion admits a direct reading. If the founder dies at its first step, which happens with probability $1 - p$, extinction has already occurred. If instead it divides, with probability p , then extinction by generation $n + 1$ requires each of the two fresh lineages to die out within its own n generations, which happens with probability u_n^2 .

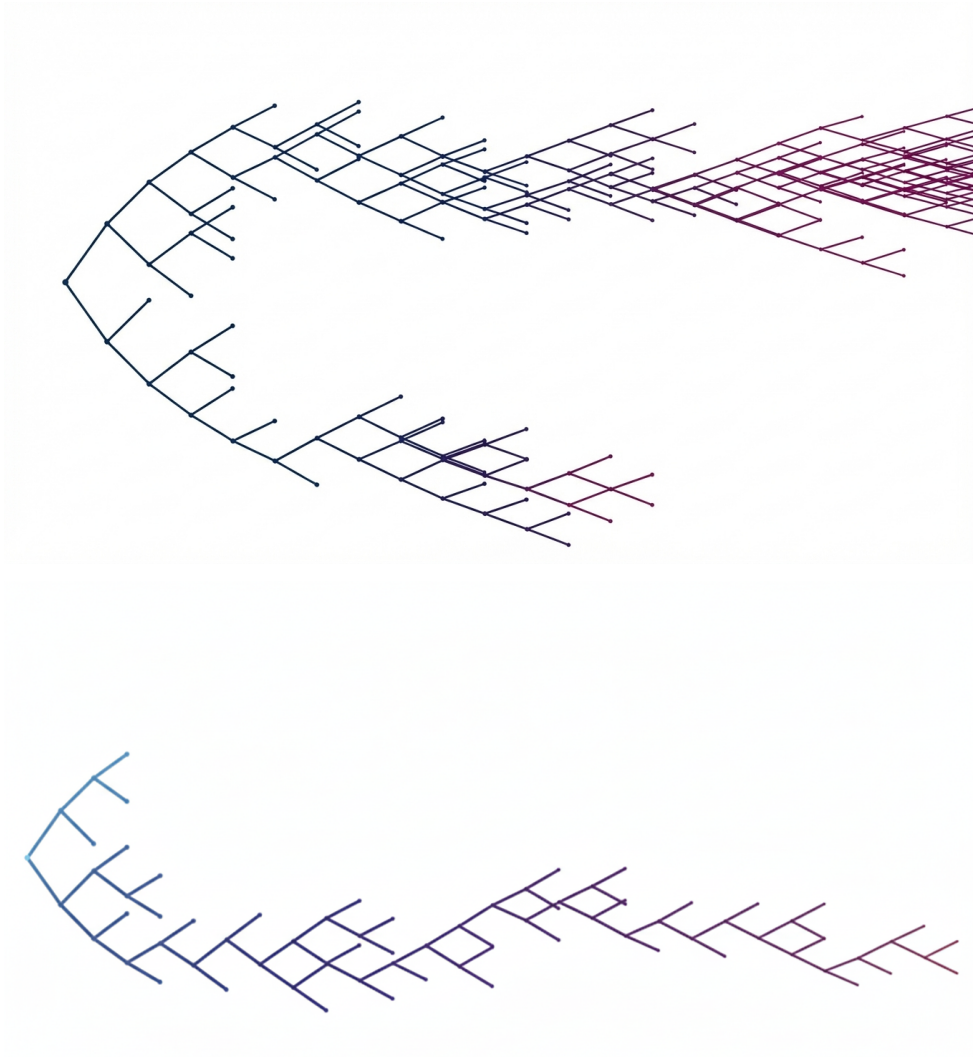


Figure 1: Realisations of the simple Galton–Watson process, drawn as genealogies with generation increasing to the right. Top: supercritical, $p = 0.58$; the number of lines grows without bound. Bottom: subcritical, $p = 0.47$; the same construction, but the tree terminates. Colour encodes generation number.

It is more convenient to work with the complement,

$$S_n = 1 - u_n = \Pr \{ \mathbf{Z}_n > 0 \},$$

the probability that the process is still alive at generation n . Substituting $u_n = 1 - S_n$ into (4) and simplifying gives

$$S_{n+1} = 2pS_n - pS_n^2, \quad S_0 = 1. \quad (5)$$

Since a population cannot return from extinction, u_n is non-decreasing and bounded above by 1; equivalently S_n is non-increasing and bounded below by 0, and therefore converges to some $S_\infty \in [0, 1]$.

2.2 Certainty of extinction

Because S_n converges, the limit must be a fixed point of (5). Setting $S_{n+1} = S_n = S_\infty$,

$$S_\infty = 2pS_\infty - pS_\infty^2 \quad \Longleftrightarrow \quad pS_\infty(S_\infty + \frac{1}{p} - 2) = 0,$$

whose roots are

$$S_\infty = 0 \quad \text{and} \quad S_\infty = 2 - \frac{1}{p} = \frac{2p-1}{p}.$$

For $p < \frac{1}{2}$ the second root is negative and so carries no probabilistic meaning; the only admissible limit is $S_\infty = 0$ and extinction is certain. At $p = \frac{1}{2}$ the two roots collide at the origin, and extinction remains certain. For $p > \frac{1}{2}$ a positive root appears, $S_\infty = (2p-1)/p$, and it is this root that is attained: the process survives forever with positive probability. Figure 2 shows the transition.

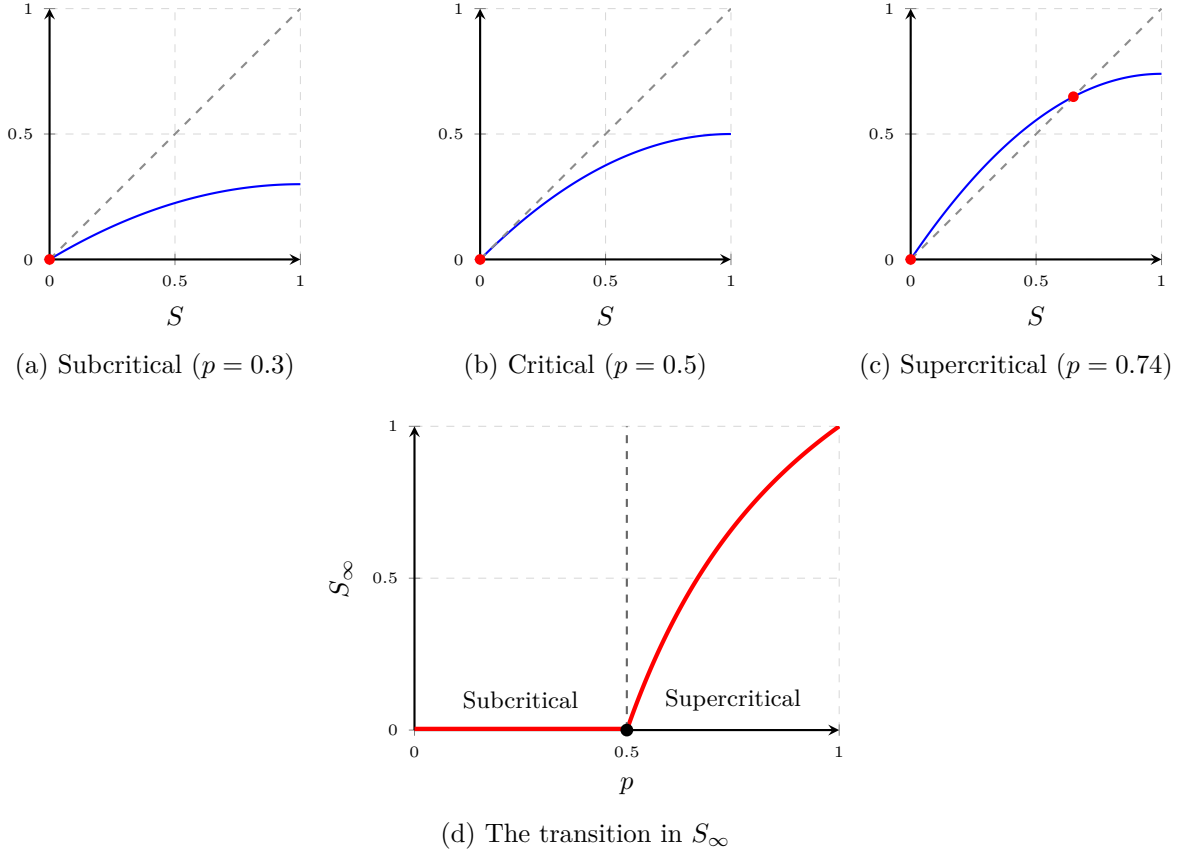


Figure 2: Steady states of the survival recursion (5). Panels (a)–(c) show the map $S \mapsto 2pS - pS^2$ (blue) against the diagonal (dashed); fixed points are the intersections, marked in red. Below and at criticality the origin is the only fixed point in $[0, 1]$, and it is attracting, so extinction is certain. At $p = \frac{1}{2}$ the map becomes tangent to the diagonal at the origin. For $p > \frac{1}{2}$ the origin turns repelling and a stable positive fixed point $S_\infty = (2p-1)/p$ separates from it, traced in panel (d). The survival probability is identically zero below the critical point and rises continuously above it.

2.3 The critical case: a power law and an infinite expected lifetime

At $p = \frac{1}{2}$ the expectation is constant, $\mathbb{E}(\mathbf{Z}_n) = 1^n = 1$ for all n , and yet extinction is certain. The reconciliation is that the expected *lifetime* is infinite: rare realisations survive for enormously long

times and grow enormously large, and they do so at exactly the rate needed to hold the mean at one. This is worth demonstrating, because the same calculation supplies the asymptotics of S_n that [section 5](#) will need.

Let

$$T = \inf\{n : \mathbf{Z}_n = 0\}$$

be the extinction time, so that the process passes through generations $0, 1, \dots, T-1$ and is dead at T . Writing T as a sum of indicators,

$$T = \sum_{n=0}^{\infty} \mathbf{1}_{\{T > n\}},$$

and taking expectations term by term (all terms are non-negative, so the exchange is legitimate),

$$\mathbb{E}(T) = \sum_{n=0}^{\infty} \Pr\{T > n\} = \sum_{n=0}^{\infty} \Pr\{\mathbf{Z}_n > 0\} = \sum_{n=0}^{\infty} S_n. \quad (6)$$

So the question becomes: how fast does S_n go to zero at criticality?

Setting $p = \frac{1}{2}$ in [\(5\)](#) gives $S_{n+1} = S_n - \frac{1}{2}S_n^2$, that is

$$S_{n+1} - S_n = -\frac{1}{2}S_n^2.$$

Once n is large the unit generational increment is small compared with the scale on which S_n varies, and the difference on the left is well approximated by a derivative:

$$\frac{dS}{dn} = -\frac{1}{2}S^2, \quad \text{with general solution} \quad S(n) = \frac{2}{n + \text{const}}.$$

Hence

$$S_n \sim \frac{2}{n} \quad (n \rightarrow \infty), \quad (7)$$

which the simulations of [figure 4\(a\)](#) confirm. Substituting into [\(6\)](#), the tail of the sum behaves like twice the harmonic series,

$$\mathbb{E}(T) = \sum_{n=0}^{\infty} S_n \sim \sum_{n=1}^{\infty} \frac{2}{n},$$

which diverges. Since every S_n is positive, $\mathbb{E}(T) = \infty$. The expected lifetime of the critical Galton–Watson process is infinite even though extinction occurs with probability one.

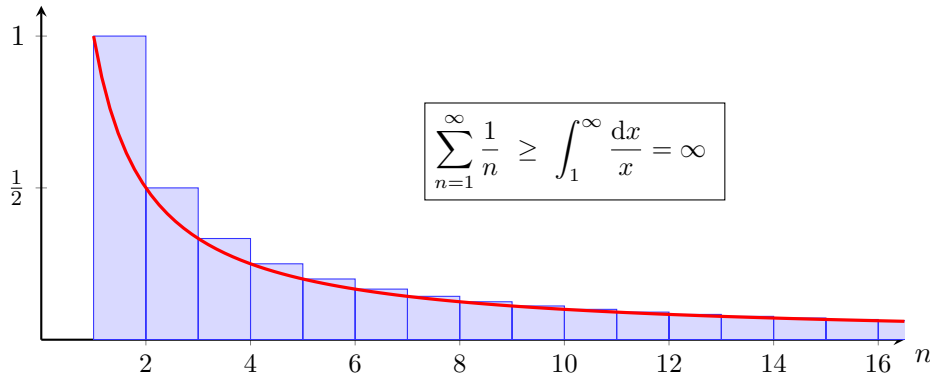


Figure 3: Divergence of the harmonic series. Each blue rectangle has area $1/n$ and contains the region under the red curve $1/x$ on $[n, n+1]$, so the sum of the areas exceeds $\int_1^{\infty} x^{-1} dx = \infty$. Because $S_n \sim 2/n$ at criticality, the same divergence carries over to $\mathbb{E}(T) = \sum_n S_n$.

The heavy tail responsible for this is visible directly in simulation. For a power-law distributed variable there is a modest but non-negligible chance of an extraordinarily large realisation, and a sample mean computed from such a variable does not settle: the larger the sample, the larger the mean, because the occasional enormous value keeps pushing the normalised sum upward. Figure 4 collects the three classical critical exponents.

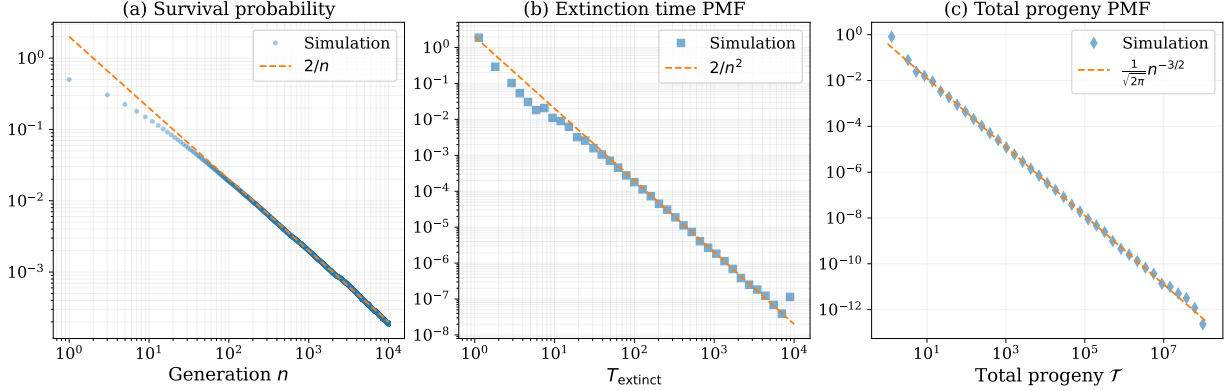


Figure 4: Power-law behaviour in the critical Galton–Watson process ($p = 0.5$), from 10^6 realisations. (a) Survival probability $S_n = \Pr\{\mathbf{Z}_n > 0\}$, showing the late-generation scaling $S_n \sim 2/n$ of (7). (b) Probability mass function of the extinction time $T = \inf\{n : \mathbf{Z}_n = 0\}$, with heavy tail $\Pr\{T = n\} \sim 2/n^2$. (c) Probability mass function of the total progeny $\mathcal{T} = \sum_{k \geq 0} \mathbf{Z}_k$, displaying the classical critical branching asymptotic $\Pr\{\mathcal{T} = n\} \sim n^{-3/2}$.

3 The limiting conditional distribution

It is often the case in stochastic processes, as it is here for $p < \frac{1}{2}$, that fixation to a particular state is inevitable. In certain cases, though, it is observed that long before absorption the process settles into a distribution that no longer changes with time — one that describes the population conditional on its not yet having been absorbed. This phenomenon is called *quasi-stationarity*, and the distribution it converges to is the quasi-stationary, or Yaglom, limit [3, 1]. This section establishes it for the binary Galton–Watson process by computing the limiting conditional mean and variance.

3.1 Late-time survival probability

Everything follows from the behaviour of S_n . Take $p < \frac{1}{2}$, so that death of an individual is more probable than its reproduction. Then $S_n \rightarrow 0$, and as it does so the quadratic term in

$$S_{n+1} = 2pS_n - pS_n^2$$

becomes negligible beside the linear one. The recursion is therefore increasingly well approximated by

$$S_{n+1} \approx 2p S_n, \tag{8}$$

and so at late times

$$S_n \sim A(p) (2p)^n \quad (n \rightarrow \infty), \quad A(p) = \lim_{n \rightarrow \infty} \frac{S_n}{(2p)^n}. \tag{9}$$

The constant $A(p)$ absorbs the cumulative effect of the quadratic term over the early generations, before the linear regime takes hold: it is what remains of the lingering conditions on S_n accumulated before attenuation to (8). The limit exists and is finite and strictly positive for $p \in [0, \frac{1}{2})$; a proof via an infinite product is given in [section 5.1](#), and [section 5.2](#) locates it in $(0, \frac{1}{2}]$.

Remark 3.1. The relation (9) is an asymptotic equivalence, not an identity. The limit of S_n itself is 0 for $p < \frac{1}{2}$, and one must not write $\lim_n S_n = A(p)(2p)^n$: the right-hand side still depends on n . What is being asserted is that the ratio $S_n/(2p)^n$ converges.

3.2 Mean

Decompose the expected population size into the contributions from surviving and extinct realisations:

$$\mathbb{E}(\mathbf{Z}_n) = \Pr\{\mathbf{Z}_n = 0\} \mathbb{E}(\mathbf{Z}_n | \mathbf{Z}_n = 0) + \Pr\{\mathbf{Z}_n > 0\} \mathbb{E}(\mathbf{Z}_n | \mathbf{Z}_n > 0).$$

The first term vanishes identically, so with $\Pr\{\mathbf{Z}_n > 0\} = S_n$,

$$\mathbb{E}(\mathbf{Z}_n | \mathbf{Z}_n > 0) = \frac{\mathbb{E}(\mathbf{Z}_n)}{S_n} = \frac{(2p)^n}{S_n}, \quad (10)$$

using $\mathbb{E}(\mathbf{Z}_n) = (2p)^n$ from (2). We know what $(2p)^n$ does; the content is all in S_n , and (9) supplies it:

$$\mathbb{E}(\mathbf{Z}_n | \mathbf{Z}_n > 0) \xrightarrow{n \rightarrow \infty} \frac{(2p)^n}{A(p)(2p)^n} = \frac{1}{A(p)}. \quad (11)$$

The geometric factors cancel exactly. At late times the surviving population has a constant expected size, and that size is the reciprocal of $A(p)$. For this reason $1/A(p)$ is referred to below as the *quasi-stationary mean*.

[Figure 5](#) shows the convergence. It also shows the difficulty that [section 5](#) will have to confront: the closer p is to $\frac{1}{2}$, the higher the plateau and the longer the process takes to reach it.

3.3 Variance

The mean is not the whole of quasi-stationarity. To show that the conditional *distribution* settles down we need at least the second moment, and computing it is a worthwhile exercise in its own right.

3.3.1 The second moment $\mathbb{E}(\mathbf{Z}_n^2)$

For a random variable $\mathbf{X}$ the moment generating function is $\mathbb{E}(e^{t\mathbf{X}})$, which can be written in terms of the probability generating function as $\phi(e^t)$. For the binary offspring law (3),

$$M_1(t) = \phi_1(e^t) = p e^{2t} + (1 - p), \quad (12)$$

the subscript 1 indicating relevance to generation $n = 1$. Moments follow from

$$\mathbb{E}(\mathbf{Z}_n^r) = \left. \frac{d^r}{dt^r} M_n(t) \right|_{t=0}. \quad (13)$$

For the second moment of $\mathbf{Z}_1$, differentiating (12) twice and evaluating at $t = 0$ gives

$$\left. \frac{d^2}{dt^2} M_1(t) \right|_{t=0} = 4p e^{2t} \Big|_{t=0} = 4p.$$

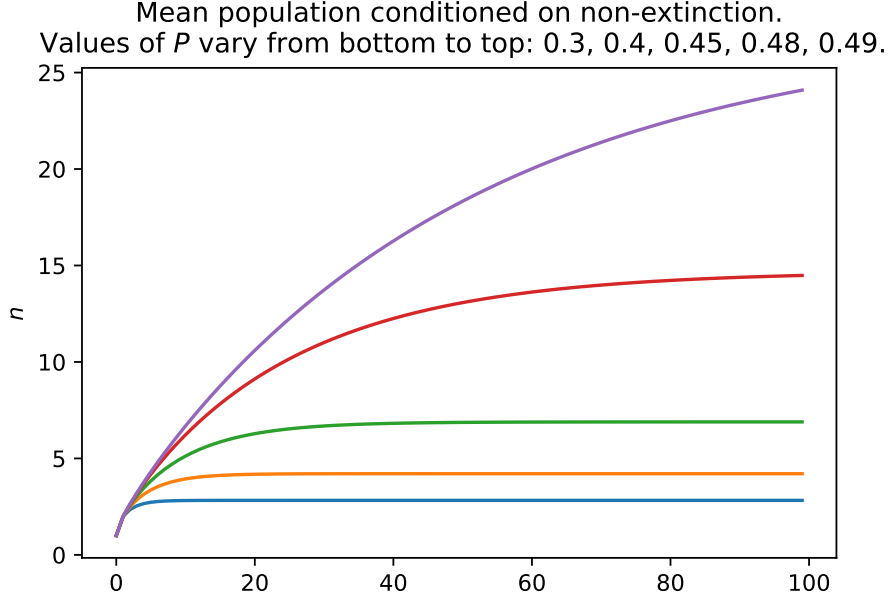


Figure 5: The conditional mean $(2p)^n/S_n$ plotted against generation number n , with S_n computed by iterating (5). From bottom to top, $p = 0.3, 0.4, 0.45, 0.48, 0.49$. Each curve approaches the constant $1/A(p)$; the values are 2.825, 4.208, 6.893, 14.703 and 27.476 respectively, computed as in section 5.7. The lowest three have essentially converged by $n = 100$, but the $p = 0.49$ curve has reached only 24.1 of its limiting 27.5 — an early sign of the near-critical slowdown discussed in section 5.7. (The vertical axis is mislabelled n in the original plotting script; it is the conditional mean.)

That is only the first generation. To reach generation n we must dig a little deeper. Recall that the n -step probability generating function is the n -fold composition

$$\phi_n(z) = \underbrace{\phi_1(\phi_1(\dots\phi_1(z)))}_{n \text{ times}},$$

so that

$$M_n(t) = \phi_1(\phi_1(\dots\phi_1(e^t))).$$

Differentiating once by the chain rule,

$$M'_n(t) = \phi'_1(\phi_{n-1}(e^t)) \cdot \phi'_1(\phi_{n-2}(e^t)) \cdots \phi'_1(\phi_1(e^t)) \cdot \phi'_1(e^t) \cdot e^t. \quad (14)$$

Now

$$\phi'_1(\phi_k(e^t)) = \frac{d}{d\phi_k(e^t)}(p[\phi_k(e^t)]^2 + (1-p)) = 2p\phi_k(e^t),$$

and $\phi'_1(e^t) = 2pe^t$, so (14) collapses to

$$M'_n(t) = (2p)\phi_{n-1}(e^t) \cdot (2p)\phi_{n-2}(e^t) \cdots (2p)\phi_1(e^t) \cdot (2p)e^{2t},$$

that is,

$$M'_n(t) = (2p)^n \cdot M_{n-1}(t) \cdot M_{n-2}(t) \cdots M_1(t) \cdot e^{2t}. \quad (15)$$

Before differentiating again, note that every factor here equals 1 at $t = 0$: $M_k(0) = 1$ for all k , and $e^0 = 1$. This is what makes the second derivative tractable. Differentiating the product (15) gives a sum of terms, each identical to (15) except that one factor has been differentiated:

$$\begin{aligned} M_n''(t) = (2p)^n & \left[M_{n-1}'(t) \cdot M_{n-2}(t) \cdots M_1(t) \cdot e^{2t} \right. \\ & + M_{n-1}(t) \cdot M_{n-2}'(t) \cdots M_1(t) \cdot e^{2t} \\ & \quad \vdots \\ & + M_{n-1}(t) \cdot M_{n-2}(t) \cdots M_1'(t) \cdot e^{2t} \\ & \left. + M_{n-1}(t) \cdot M_{n-2}(t) \cdots M_1(t) \cdot 2e^{2t} \right]. \end{aligned}$$

At $t = 0$ all the undifferentiated factors become 1, leaving

$$M_n''(0) = (2p)^n [M_{n-1}'(0) + M_{n-2}'(0) + \cdots + M_1'(0) + 2].$$

But $M_k'(0)$ is just the mean population at generation k , namely $(2p)^k$, so

$$M_n''(0) = (2p)^n [(2p)^{n-1} + (2p)^{n-2} + \cdots + (2p) + 2] = (2p)^n \left[\sum_{i=0}^{n-1} (2p)^i + 1 \right].$$

Summing the finite geometric series,

$$\mathbb{E}(\mathbf{Z}_n^2) = M_n''(0) = (2p)^n \left[\frac{1 - (2p)^n}{1 - 2p} + 1 \right],$$

and rearranging,

$$\mathbb{E}(\mathbf{Z}_n^2) = (2p)^n \left(1 + \frac{1}{1 - 2p} \right) - (2p)^{2n} \left(\frac{1}{1 - 2p} \right). \quad (16)$$

The minus sign is required: at $n = 1$ the expression must return $\mathbb{E}(\mathbf{Z}_1^2) = 4p$, which it does.

3.3.2 Late-time variance

From $\text{Var}(\mathbf{Z}_n) = \mathbb{E}(\mathbf{Z}_n^2) - \mathbb{E}(\mathbf{Z}_n)^2$ and (16),

$$\text{Var}(\mathbf{Z}_n) = (2p)^n \left(1 + \frac{1}{1 - 2p} \right) - (2p)^{2n} \left(\frac{1}{1 - 2p} \right) - (2p)^{2n}.$$

For $2p < 1$ the terms in $(2p)^{2n}$ are negligible beside $(2p)^n$ at large n , so

$$\text{Var}(\mathbf{Z}_n) \rightarrow (2p)^n \left(1 + \frac{1}{1 - 2p} \right).$$

It is tempting to read stationarity of the conditional variance straight off this, but that would be too quick: the conditional variance is not the unconditional one divided by anything simple. Start instead from the conditional moments. Using (9),

$$\mathbb{E}(\mathbf{Z}_n^2 \mid \mathbf{Z}_n \neq 0) = \frac{\mathbb{E}(\mathbf{Z}_n^2)}{S_n} \xrightarrow{n \rightarrow \infty} \frac{(2p)^n \left(1 + \frac{1}{1 - 2p} \right)}{A(2p)^n} = \frac{1}{A} \left(1 + \frac{1}{1 - 2p} \right),$$

$$\mathbb{E}(\mathbf{Z}_n \mid \mathbf{Z}_n \neq 0) = \frac{\mathbb{E}(\mathbf{Z}_n)}{S_n} \xrightarrow{n \rightarrow \infty} \frac{(2p)^n}{A(2p)^n} = \frac{1}{A}.$$

Both limits are finite, so the conditional variance does indeed tend to a constant:

$$\text{Var}(\mathbf{Z}_n \mid \mathbf{Z}_n \neq 0) \longrightarrow \frac{1}{A} \left(1 + \frac{1}{1-2p} \right) - \frac{1}{A^2}. \quad (17)$$

The first two moments of the conditional population therefore both stabilise, and the same argument extended to higher moments gives the full quasi-stationary distribution. Note that (17), like (11), is expressed entirely in terms of p and the single unknown constant $A(p)$. Everything now turns on that constant.

3.4 A continuum view of the survival recursion

Before turning to $A(p)$ proper it is useful to record a continuum approximation that recurs later. Writing $r = 1 - 2p$ for the distance from criticality, the recursion (5) rearranges to

$$S_{n+1} - S_n = -rS_n - pS_n^2. \quad (18)$$

For large n the unit increment is small on the scale over which S varies, and the left-hand side is well approximated by a derivative, giving the Riccati equation

$$\frac{dS}{dn} = -rS - pS^2. \quad (19)$$

Substituting $u = 1/S$ linearises it, $u' = ru + p$, with solution $u(n) = Ke^{rn} - p/r$ and hence

$$S(n) = \frac{r}{Ke^{rn} - p}, \quad (20)$$

K a constant fixed by the initial condition. Setting $r = 0$ recovers the critical scaling $S \sim 2/n$ of (7); keeping $r > 0$ recovers the geometric decay $S \sim A(2p)^n$ of (9), since $(2p)^n = e^{n \ln(1-r)}$ agrees with e^{-rn} to leading order in r . The constant K is exactly the object that the continuum approximation cannot supply, because it is determined by the early generations where the approximation is invalid — which is another way of saying that it is $A(p)$, and that no argument of this kind will produce it. [Section 5.3](#) shows how much can nevertheless be extracted near criticality.

4 Survival of small populations

Consider again the basic Galton–Watson process in which each cell leaves behind two daughters or dies, and suppose it is near critical — either just below or just above. With the probability of replication p and of death $1 - p$ each close to a half, a lineage dies out at its first round almost half the time. Should it pass one generation, both of its two offspring will fail at their own first attempt with probability around 0.25. So the lineage of that initial cell reaches the second generation only with likelihood approximately 0.375:

$$0.375 \approx \underbrace{(1 - 0.5)}_{\text{no extinction at the first step}} \times \underbrace{(1 - 0.5 \times 0.5)}_{\text{nor at the second}}.$$

More generally, survival to a given generation is quantified by iterating the probability generating function. With $\phi(z) = pz^2 + (1 - p)$ from (3), the extinction and survival probabilities follow from

$$u_{n+1} = pu_n^2 + 1 - p, \quad S_{n+1} = 2pS_n - pS_n^2, \quad (21)$$

started at $u_0 = 0$, $S_0 = 1$. At exactly critical branching this gives, for $n = 1, 2, 3, 4$,

$$u_n = 0.500, 0.625, 0.695, 0.741 \quad (3 \text{ s.f.}),$$

so the likelihood of process survival decays quickly with ongoing rounds of replication, and does so almost as fast just above criticality as just below (figure 6).

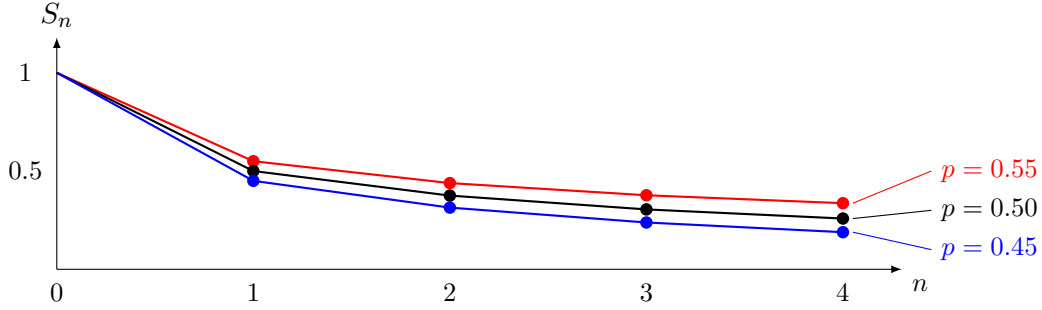


Figure 6: Survival probability over the first four generations from a single progenitor, from (21). Even in the supercritical case ($p = 0.55$) the early decay is steep, and the three curves have barely separated by $n = 4$. Whatever advantage supercriticality confers, it is not an early-generation advantage.

4.1 An initial cohort of size k

Now suppose $\mathbf{Z}_0 = k$: the process begins not with a single unit but with a group, small to moderate in number, perhaps $k = 10$. Under supercritical branching, as the count increases the growth becomes more predictably exponential — a large number of units lets the expected dynamics show through the haze of individual stochasticity. But the founding cohort is precisely the regime in which that haze dominates.

So for such an initial colony — say ten pathogenic bacteria in a mammalian host — what are its chances of persistence, of going on to found a full-blown infection? The question is answered by asking for the probability that at least one descendant remains, that is, that not every one of the k founding lineages has gone extinct. Since the lineages are independent and each dies out by generation n with probability ${}_1u_n$, all k disappear with probability $({}_1u_n)^k$, and so

$$S_n^{(k)} = 1 - ({}_1u_n)^k. \quad (22)$$

Figures 7 and 8 show $S_n^{(k)}$ through 500 generations for $k = 1, 10, 100$ and 1000. The effect visible in figure 7 — that a greater first cohort gives improved longevity — is highly sensitive to the criticality of the process. Persistence improves quickly as p approaches $\frac{1}{2}$ from below and continues to improve dramatically into supercriticality; figure 8 shows the same experiment with p slightly above and slightly below $\frac{1}{2}$.

4.2 An aside on early replication

It is worth noting that the same conditioning effect has been identified in a quite different setting: the longevity and size of phylogenetic clades. Those clades that have persisted through the ages, and which are often of superior diversity, turn out to have experienced an above-average early rate of

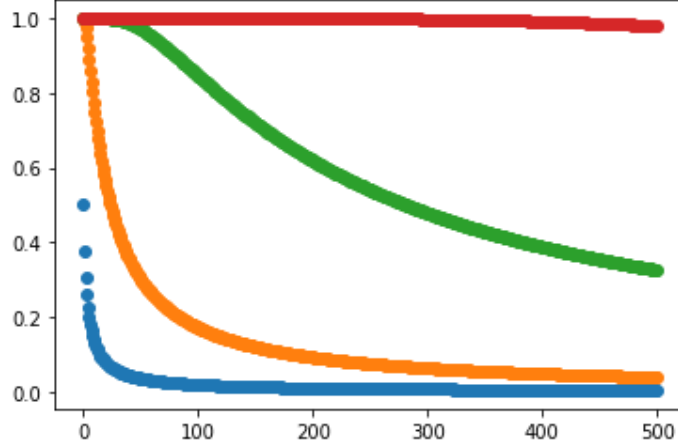


Figure 7: Survival probability $S_n^{(k)}$ from (22) out to 500 generations at exact criticality, $p = 0.5$, for $k = 1, 10, 100, 1000$ (blue through to red). A larger founding cohort buys longevity, but at criticality every curve is still headed for zero.

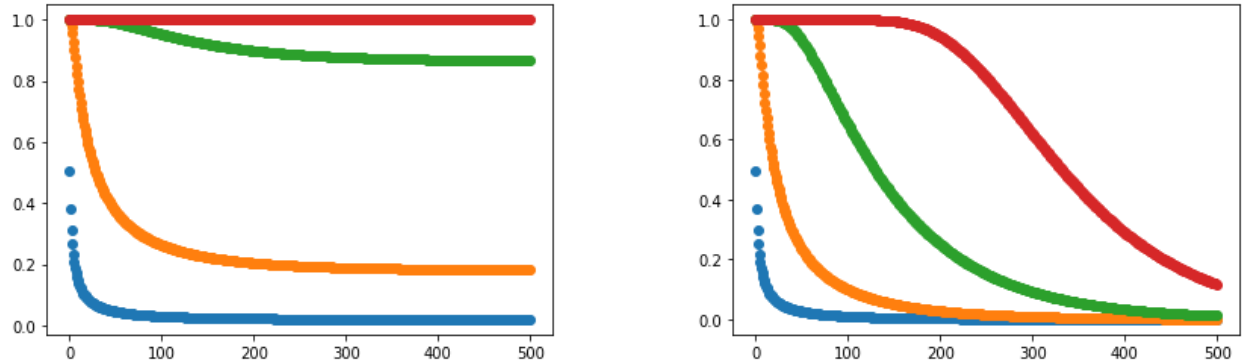


Figure 8: As figure 7, but with p displaced slightly from criticality: $p = 0.505$ on the left (supercritical), $p = 0.495$ on the right (subcritical). The benefit of a large initial cohort is acutely sensitive to which side of $\frac{1}{2}$ the process sits on; a displacement of half a percent is enough to separate curves that were nearly coincident in figure 7.

species origination — an observation drawn from the fossil record itself. Budd and Mann [4] argue that this empirical fact may be a statistical consequence rather than a biological one: if one runs a large number of homogeneous birth–death processes and retains only those which happen to survive to some arbitrary large positive time — “the present” — then the retained set is automatically enriched for early rapid origination, with no change in the underlying rates required. The apparent slowdown in diversification that follows is then not a slowdown at all but a reversion to the mean, an effect known as the *push of the past*.

That is precisely the conditioning of section 3 seen from the other side. There we conditioned a subcritical process on survival and found that its mean settled to $1/A(p)$ rather than decaying; here one conditions a critical-or-supercritical process on survival to the present and finds its early growth inflated. In both cases the conditioning, not the mechanism, does the work.

The parallel suggests a hypothesis on the pathogenic side: that it is of advantage to an infectious agent to secure a high probability of early replication, even at some cost to individual survival

through later generations. [Figure 6](#) is the reason to take the suggestion seriously — the first few generations are where nearly all of the extinction risk is concentrated.

4.3 Discrete birth–death–catastrophe

Until the point of catastrophe, a discrete-time birth–death–catastrophe process behaves exactly as the basic Galton–Watson model with regard to its limiting conditional distribution. Conditioning on non-rupture, and conditioning also on non-extinction by death, the process exhibits a stationary limiting distribution. Whether a quasi-stationary distribution survives when one conditions on *neither* death nor catastrophe is a separate question, and one best settled by simulation. Either way it is useful to have the probability of non-catastrophe in hand.

Probability of rupture at step n

Let κ be the probability that a given individual initiates catastrophe at a step. A single individual fails to cause rupture with probability $1 - \kappa$, two with $(1 - \kappa)^2$, and in general a cohort of size N with $(1 - \kappa)^N$.

Consider first the case of birth and catastrophe only, with no death. Then the census of the living population at any generation is known with certainty: from a single progenitor the population simply doubles at each step prior to catastrophe. The process therefore comes to its abrupt halt in generation n with probability

$$1 - (1 - \kappa)^{2^{n-1}},$$

that is, one minus the probability that none of the 2^{n-1} individuals present caused it.

Probability of process survival

Along the same lines we can define, still for the no-death case, the probability S_n that the process survives to generation n . Survival to generation 0 is certain, $S_0 = 1$. Survival to generation 1 requires only that the founder replicated without rupture, $S_1 = 1 - \kappa$. For persistence to step 2 not only must the founder divide, but so must each of its two offspring:

$$S_2 = (1 - \kappa)(1 - \kappa)^2.$$

The pattern is clear. Survival to the n th generation requires every prior step to have been passed, so

$$S_n = \prod_{i=0}^{n-1} (1 - \kappa)^{2^i} = (1 - \kappa)^{\sum_{i=0}^{n-1} 2^i} = (1 - \kappa)^{2^n - 1}. \quad (23)$$

Death included

The difficulty comes in defining S_n once death is allowed, because the size of the living population is then no longer known with certainty from one generation to the next. The natural guess is that the probabilities somehow shake out when the expected population is used in place of the actual one:

$$S_n \stackrel{?}{=} (1 - \kappa)^{\sum_{i=0}^{n-1} \mathbb{E}(\mathbf{X}_i)}. \quad (24)$$

This is almost certainly not an identity, and it is worth being explicit about why. The map $N \mapsto (1 - \kappa)^N$ is convex, so by Jensen’s inequality $\mathbb{E}((1 - \kappa)^{\mathbf{X}_i}) \geq (1 - \kappa)^{\mathbb{E}(\mathbf{X}_i)}$: replacing the

random census by its mean systematically underestimates the chance of getting through a step. Nor are the steps independent, since surviving one is evidence of a smaller population, which raises the chance of surviving the next. Both effects push the same way, so (24) is better read as a lower bound than as an equality. How tight a bound it is remains worth checking numerically.

4.4 Continuous-time birth–death

The discrete process resists a closed form for $A(p)$, as section 5 will show at length. Its continuous-time counterpart does not, and the comparison between the two turns out to be informative, so we record the continuous-time result here.

For the simple linear birth–death process with per-capita birth rate λ and death rate μ , started from N individuals, the extinction probability is [5]

$$p_0(t) = \begin{cases} \left(\frac{\mu - \mu e^{(\mu-\lambda)t}}{\lambda - \mu e^{(\mu-\lambda)t}} \right)^N, & \lambda \neq \mu, \\ \left(\frac{\lambda t}{1 + \lambda t} \right)^N, & \lambda = \mu, \end{cases} \quad (25)$$

from which the survival probability is $S^{(N)}(t) = 1 - p_0(t)$.

Limiting mean conditioned on non-extinction

The simple birth–death model has a stationary conditional mean when $\mu > \lambda$, that is, in the subcritical case. Exactly as in section 3.2,

$$\mathbb{E}(\mathbf{X}_t \mid \mathbf{X}_t > 0) = \frac{\mathbb{E}(\mathbf{X}_t)}{S^{(N)}(t)}, \quad (26)$$

and the late-time limit follows on applying L'Hôpital's rule. It is quicker to take $N = 1$ and compute directly. With $\mathbb{E}(\mathbf{X}_t) = e^{(\lambda-\mu)t}$ and

$$S(t) = 1 - \left(\frac{\mu - \mu e^{(\mu-\lambda)t}}{\lambda - \mu e^{(\mu-\lambda)t}} \right) = \frac{\lambda - \mu}{\lambda - \mu e^{(\mu-\lambda)t}},$$

(26) gives

$$\mathbb{E}(\mathbf{X}_t \mid \mathbf{X}_t > 0) = \frac{\lambda e^{(\lambda-\mu)t} - \mu}{\lambda - \mu}.$$

For $\mu > \lambda$ the exponential decays and the limiting conditional mean is a constant,

$$\lim_{t \rightarrow \infty} \mathbb{E}(\mathbf{X}_t \mid \mathbf{X}_t > 0) = \frac{\mu}{\mu - \lambda} = \frac{1}{1 - \lambda/\mu} \left(= \frac{1}{A_c} \right). \quad (27)$$

Written in terms of the replication probability $p = \lambda/(\lambda + \mu)$, so that $\lambda/\mu = p/(1 - p)$,

$$\frac{1}{A_c} = \frac{1 - p}{1 - 2p}, \quad \text{that is} \quad A_c(p) = \frac{1 - 2p}{1 - p}. \quad (28)$$

The subscript “c” marks this as the continuous-time constant, to distinguish it from the discrete-time $A(p)$ of (9).

Figure 9 plots $A_c(p)$ against numerically obtained values of the discrete-time $A(p)$, and raises an obvious question: why does the continuous curve start at 1 and the discrete one at $\frac{1}{2}$? Merely halving A_c does not make the two coincide, because at the other end of the range they meet. The answer is given in section 5.4, once the necessary machinery is in place; it turns out that $A_c(p)/2$ is an exact lower bound for $A(p)$, attained only in the limit $p \rightarrow 0$.

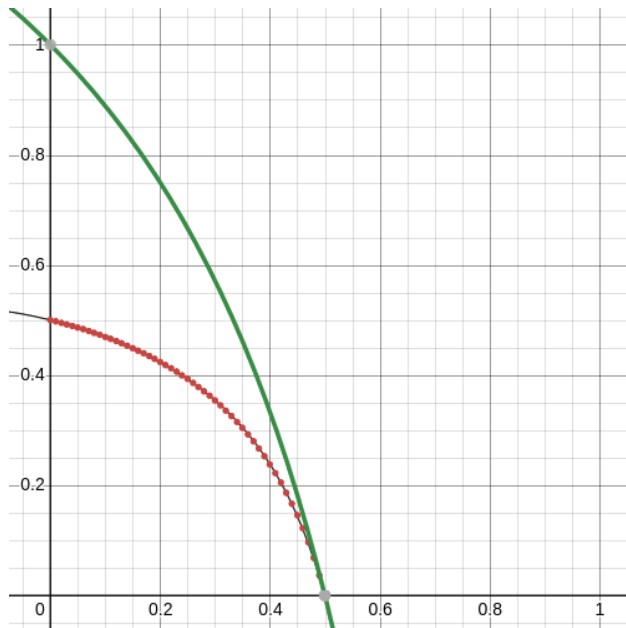


Figure 9: The continuous-time constant $A_c(p) = (1 - 2p)/(1 - p)$ (green) against high-precision numerical values of the discrete-time constant $A(p)$ (red points). Both vanish at $p = \frac{1}{2}$, but at $p = 0$ the continuous curve starts at 1 and the discrete one at $\frac{1}{2}$, and the two are not related by any constant factor in between. The black line is a formula obtained by symbolic regression (section 5.5); it tracks the points closely but is not exact. Section 5.4 accounts for the discrepancy exactly.

5 The constant $A(p)$

Everything in section 3 reduced to the single constant

$$A(p) = \lim_{n \rightarrow \infty} \frac{S_n}{(2p)^n}, \quad p \in [0, \frac{1}{2}), \quad (29)$$

whose reciprocal is the quasi-stationary mean. Approximate values are easy enough to obtain: iterate $S_{n+1} = f(S_n)$ until S_n is very nearly stationary, then form $S_n/(2p)^n$. Repeating over a range of p traces the curve of figure 9, and this works well for most of the interval. It fails as $p \rightarrow \frac{1}{2}^-$, because the number of generations required for S_n to reach near-stationarity grows without bound.

No formula in elementary functions is known for $A(p)$, and a good deal of effort went into looking for one. This section sets out what was found: an exact infinite product (section 5.1); an exact series with two-sided bounds that pin $A(p)$ between the continuous-time constant of section 4.4 and twice it (section 5.2); the near-critical asymptotic $A(p) \sim 2(1 - 2p)$ (section 5.3); a resolution of the discrepancy between the discrete and continuous curves (section 5.4); an account of the symbolic-regression searches (section 5.5); and finally the dynamical argument that explains why the search was doomed (section 5.6). Section 5.7 draws the practical conclusions.

5.1 An infinite-product representation

Take the nonlinear map for the survival probability,

$$S_{n+1} = 2pS_n - pS_n^2, \quad S_0 = 1,$$

and substitute $S_n = 2w_n$. This gives

$$w_{n+1} = 2p w_n (1 - w_n) = r w_n (1 - w_n), \quad r = 2p, \quad w_0 = \frac{1}{2}, \quad (30)$$

which is the logistic map — the same map that exhibits the celebrated period-doubling route to chaos as r increases from 0 to 4 ([figure 10](#)). Our subcritical window is $r \in [0, 1)$, at the extreme left of that picture, where the origin is a globally attracting fixed point. That this innocuous corner of the logistic family is nevertheless enough to obstruct a closed form is the substance of [section 5.6](#).

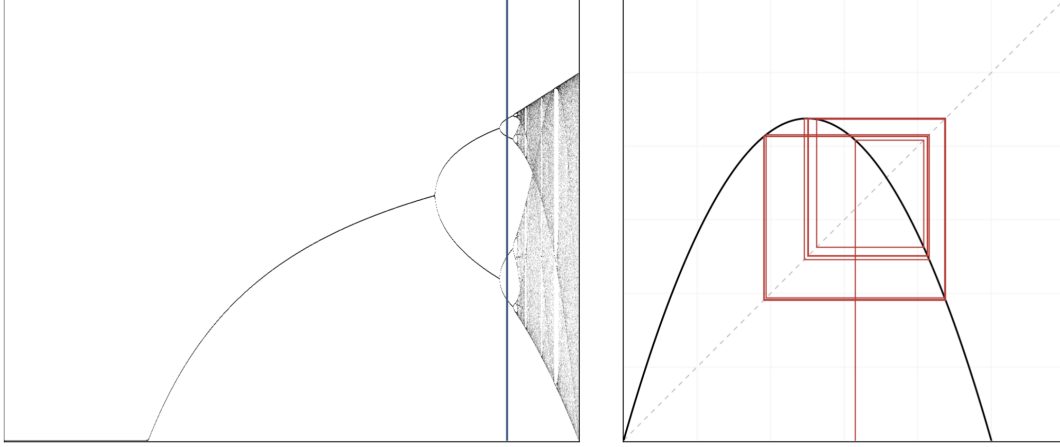


Figure 10: Bifurcation diagram of the logistic map $w \mapsto rw(1 - w)$, showing the period doubling of stable states and the onset of chaos. The blue sample line on the left is at $r = 3.5255$; the right-hand plot shows a trajectory of the map at the same r . The regime relevant to subcritical branching is $r = 2p \in [0, 1)$, off the left-hand edge of this diagram, where the dynamics are as simple as they can be.

In terms of w , [\(29\)](#) reads

$$A(p) = \lim_{n \rightarrow \infty} \frac{S_n}{(2p)^n} = 2 \lim_{n \rightarrow \infty} \frac{w_n}{r^n}. \quad (31)$$

Now any sequence can be written as a telescoping product,

$$x_n = x_0 \left(\frac{x_1}{x_0} \right) \left(\frac{x_2}{x_1} \right) \cdots \left(\frac{x_n}{x_{n-1}} \right) = x_0 \prod_{k=0}^{n-1} \frac{x_{k+1}}{x_k}, \quad \text{so} \quad \lim_{n \rightarrow \infty} x_n = x_0 \prod_{k=0}^{\infty} \frac{x_{k+1}}{x_k}.$$

Apply this with $x_n = w_n/r^n$. The ratio simplifies beautifully:

$$\frac{x_{k+1}}{x_k} = \frac{w_{k+1}/r^{k+1}}{w_k/r^k} = \frac{w_{k+1}}{w_k} \cdot \frac{1}{r} = \frac{rw_k(1 - w_k)}{w_k} \cdot \frac{1}{r} = 1 - w_k.$$

Hence, with $x_0 = w_0 = \frac{1}{2}$ and $A(p) = 2 \lim_n x_n$,

$$A(p) = 2 \cdot \frac{1}{2} \prod_{k=0}^{\infty} (1 - w_k) = \prod_{k=0}^{\infty} (1 - w_k),$$

and pulling out the $k = 0$ factor, $1 - w_0 = \frac{1}{2}$,

$$A(p) = \frac{1}{2} \prod_{k=1}^{\infty} (1 - w_k) = \frac{1}{2} \prod_{k=1}^{\infty} \left(1 - \frac{S_k}{2} \right). \quad (32)$$

Convergence

For $0 < w_k < 1$ the infinite product $\prod_{k \geq 1} (1 - w_k)$ converges to a strictly positive limit if and only if $\sum_{k \geq 1} w_k < \infty$. From (30) with $0 < r < 1$ and $0 < w_n < 1$ we have $w_{n+1} < r w_n$, so by induction $w_n \leq w_0 r^n$. Since $0 < r < 1$ this bound is summable, so $\sum_k w_k < \infty$ and the product converges to a positive finite value. This also establishes what was asserted in section 3.1: the limit (29) defining $A(p)$ exists and lies strictly between 0 and $\frac{1}{2}$ for $p \in (0, \frac{1}{2})$.

Is this an advance?

Only a modest one. (32) is exact and its partial products are monotone decreasing, so truncating gives a rigorous upper bound on $A(p)$ at every order — which the raw ratio $S_n/(2p)^n$ does not. But computing the factors still requires iterating the nonlinear map to generate the S_k , so the near-critical difficulty is untouched. What the product does supply is a route to the sharper result of the next subsection.

5.2 An exact series and two-sided bounds

The product can be converted into a telescoping sum, and the sum turns out to be much more informative. Write $r = 2p$ and note that the recursion factorises as

$$S_{n+1} = r S_n \left(1 - \frac{S_n}{2} \right).$$

Define

$$v_n = \frac{r^n}{S_n},$$

so that, by (29), $v_n \rightarrow 1/A(p)$. Then

$$v_{n+1} - v_n = \frac{r^{n+1}}{r S_n \left(1 - \frac{S_n}{2} \right)} - \frac{r^n}{S_n} = \frac{r^n}{S_n} \left[\frac{1}{1 - \frac{S_n}{2}} - 1 \right] = \frac{r^n}{S_n} \cdot \frac{S_n/2}{1 - \frac{S_n}{2}} = \frac{r^n}{2 - S_n}.$$

Summing from $v_0 = 1/S_0 = 1$ gives an exact representation of the quasi-stationary mean.

Proposition 5.1 (Series representation). *For $p \in [0, \frac{1}{2})$, with S_n generated by (5),*

$$\frac{1}{A(p)} = 1 + \sum_{n=0}^{\infty} \frac{(2p)^n}{2 - S_n}. \quad (33)$$

Unlike the product, this representation isolates the geometric decay in a factor that does not depend on the dynamics at all; the whole of the nonlinearity sits in the bounded denominator $2 - S_n$. That is what makes it useful, because S_n is confined to $(0, 1]$ and therefore $2 - S_n$ is confined to $[1, 2)$. Bounding the denominator by its extremes and summing the resulting geometric series gives the following.

Proposition 5.2 (Two-sided bounds). *Write $\varepsilon = 1 - 2p$ for the distance from criticality. Then for $p \in (0, \frac{1}{2})$*

$$\frac{\varepsilon}{1 + \varepsilon} < A(p) < \frac{2\varepsilon}{1 + 2\varepsilon}, \quad (34)$$

that is,

$$\frac{1-2p}{2(1-p)} < A(p) < \frac{2(1-2p)}{3-4p}. \quad (35)$$

The lower bound is exactly half the continuous-time constant of (28):

$$A(p) > \frac{1}{2}A_c(p). \quad (36)$$

Proof. Since $0 < S_n \leq 1$ for all n , we have $1 \leq 2 - S_n < 2$ and hence $\frac{1}{2}r^n < r^n/(2 - S_n) \leq r^n$, with the right-hand equality holding only at $n = 0$. Summing over $n \geq 0$ and using $\sum_n r^n = 1/(1 - r) = 1/\varepsilon$,

$$1 + \frac{1}{2\varepsilon} < \frac{1}{A(p)} < 1 + \frac{1}{\varepsilon},$$

and inverting gives (34). Substituting $\varepsilon = 1 - 2p$ gives (35), whose lower bound is $(1 - 2p)/(2(1 - p)) = \frac{1}{2}A_c(p)$ by (28). $\square$

Two features of [proposition 5.2](#) deserve comment, because between them they account for the whole shape of the discrete curve.

At the far subcritical end, $p \rightarrow 0$, we have $\varepsilon \rightarrow 1$ and the lower bound tends to $\frac{1}{2}$. Since $A(p) \leq \frac{1}{2}$ — see [proposition 5.3](#) below — the bound is asymptotically attained, and $A(0^+) = \frac{1}{2}$. At the critical end, $p \rightarrow \frac{1}{2}^-$, we have $\varepsilon \rightarrow 0$, the lower bound behaves like ε and the upper like 2ε ; [section 5.3](#) shows that it is the upper bound that is attained there. So $A(p)$ runs from its lower bound at one end of the interval to its upper bound at the other, which is why no constant rescaling of A_c can reproduce it.

Proposition 5.3 (Parity bound). *For the binary Galton–Watson process started from a single individual, $A(p) \leq \frac{1}{2}$ for all $p \in [0, \frac{1}{2})$, with equality approached as $p \rightarrow 0$.*

Proof. Every individual leaves either 0 or 2 offspring, so $\mathbf{Z}_n$ is even for every $n \geq 1$. Conditional on survival we therefore have $\mathbf{Z}_n \geq 2$, and hence $\mathbb{E}(\mathbf{Z}_n \mid \mathbf{Z}_n > 0) \geq 2$ for all $n \geq 1$. Taking $n \rightarrow \infty$ and using (11), $1/A(p) \geq 2$. As $p \rightarrow 0$ the conditional population is 2 with probability tending to one, so the bound is attained in the limit. $\square$

5.3 Behaviour as $p \rightarrow \frac{1}{2}^-$

The series (33) also delivers the near-critical asymptotic, and does so more transparently than the matching argument one is first tempted to try.

Fix $\varepsilon = 1 - 2p$ small. The summand $r^n/(2 - S_n)$ involves two decays on very different scales. The geometric factor $r^n = (1 - \varepsilon)^n$ decays on the scale $n \sim 1/\varepsilon$, which is large. The survival probability S_n , by contrast, is close to its critical behaviour $S_n \sim 2/n$ over that whole range, and so has already fallen to near zero after $O(1)$ generations. For the overwhelming majority of the terms that carry weight, therefore, $S_n \approx 0$ and the summand is $\approx r^n/2$. Summing,

$$\sum_{n=0}^{\infty} \frac{r^n}{2 - S_n} \approx \frac{1}{2} \sum_{n=0}^{\infty} r^n = \frac{1}{2\varepsilon},$$

so $1/A(p) \sim 1/(2\varepsilon)$ and

$$\boxed{A(p) \sim 2\varepsilon = 2(1 - 2p) \quad \text{as } p \rightarrow \frac{1}{2}^-} \quad (37)$$

Table 1: The constant $A(p)$ computed from the product (32) in 60-digit arithmetic, against the bounds of proposition 5.2 and the near-critical asymptotic $2(1 - 2p)$ of section 5.3. The bounds are loose in the middle of the range and tighten at both ends: at $p = 0.05$ the lower bound is already within 3% of A , whereas at $p = 0.4999$ it is the upper bound that is within 0.1%. The last two columns give the quasi-stationary mean and the number of factors of (32) needed for the product to reach full precision — the cost that section 5.7 has to manage.

p	$\frac{1}{2}A_c(p)$	$A(p)$	$\frac{2(1-2p)}{3-4p}$	$2(1-2p)$	$1/A(p)$	iterations
0.05	0.473684	0.486180	0.642857	1.800000	2.0568	45
0.1	0.444444	0.469382	0.615385	1.600000	2.1305	64
0.2	0.375000	0.423895	0.545455	1.200000	2.3591	113
0.3	0.285714	0.353967	0.444444	0.800000	2.8251	201
0.4	0.166667	0.237647	0.285714	0.400000	4.2079	458
0.45	0.090909	0.145069	0.166667	0.200000	6.8933	966
0.48	0.038462	0.068011	0.074074	0.080000	14.7036	2 473
0.49	0.019608	0.036395	0.038462	0.040000	27.4764	4 965
0.499	0.001996	0.003944	0.003984	0.004000	253.519	48 992
0.4999	0.000200	0.000399	0.000400	0.000400	2504.65	478 905

Equivalently $1/A(p) \sim 1/(2(1 - 2p))$: the quasi-stationary mean diverges like the reciprocal of the distance from criticality. In particular $A(p)$ meets the axis at $p = \frac{1}{2}$ with gradient -4 , a fact used repeatedly below as a test of candidate formulae.

The same result can be reached through the continuum equation of section 3.4. With $\varepsilon = 1 - 2p$ the exact difference is

$$S_{n+1} - S_n = (2p - 1)S_n - pS_n^2 = -\varepsilon S_n - \frac{1}{2}S_n^2 + \frac{\varepsilon}{2}S_n^2,$$

and dropping the $O(\varepsilon S_n^2)$ term gives $dS/dn = -\varepsilon S - \frac{1}{2}S^2$. Setting $u = 1/S$ linearises this to $u' = \varepsilon u + \frac{1}{2}$, with solution

$$u(n) = Ce^{\varepsilon n} - \frac{1}{2\varepsilon}, \quad S(n) = \frac{1}{Ce^{\varepsilon n} - \frac{1}{2\varepsilon}}.$$

Matching $S_n \sim Ae^{-\varepsilon n}$ for small ε identifies $A = 1/C$, and matching to the critical decay $S_n \sim 2/n$ in the appropriate double limit fixes $C \sim 1/(2\varepsilon)$, recovering (37). The series argument is preferable because it does not require the second matching step, which is the delicate one.

Remark 5.4 (Rate of approach). Numerically, $A(p)/2\varepsilon$ approaches 1 rather slowly. Evaluating (32) in high precision gives $A/2\varepsilon = 0.90987, 0.98612, 0.99814, 0.99977$ and 0.999972 at $\varepsilon = 2 \times 10^{-2}$ down to 2×10^{-6} . Fitting the deficit gives $1 - A/2\varepsilon \approx \varepsilon(\ln(1/\varepsilon) + c)$ with $c \approx 0.78$, so

$$A(p) = 2\varepsilon \left[1 + O(\varepsilon \ln(1/\varepsilon)) \right].$$

The logarithmic correction is why (37) is a good practical surrogate only very close to criticality, and it is worth keeping in mind when section 5.7 chooses the crossover point.

5.4 Why the discrete and continuous constants differ

We can now settle the question raised by [figure 9](#). The two constants are

$$A_c(p) = \frac{1-2p}{1-p} \quad (\text{continuous time}), \quad A(p) = \lim_{n \rightarrow \infty} \frac{S_n}{(2p)^n} \quad (\text{discrete time}),$$

and the observation was that the first starts at 1 where the second starts at $\frac{1}{2}$, yet halving the first does not superpose the curves.

[Proposition 5.2](#) explains this completely. What holds is not $A = \frac{1}{2}A_c$ but the strict inequality $A > \frac{1}{2}A_c$, and the inequality is attained only in the limit $p \rightarrow 0$. Combining with [\(28\)](#) and [\(37\)](#), the ratio runs monotonically across the interval:

$$\frac{A(p)}{A_c(p)} \rightarrow \frac{1}{2} \quad (p \rightarrow 0), \quad \frac{A(p)}{A_c(p)} \rightarrow 1 \quad (p \rightarrow \frac{1}{2}^-), \quad (38)$$

taking the values 0.503, 0.528, 0.619, 0.798, 0.928 and 0.988 at $p = 0.01, 0.1, 0.3, 0.45, 0.49, 0.499$. No constant factor will do, because the ratio is not constant.

The two endpoints have clean interpretations, and they are different in kind.

At $p \rightarrow 0$ the factor of two is a counting effect. In the binary discrete model an individual leaves either nothing or two offspring, so the surviving population is always even ([proposition 5.3](#)); deep in the subcritical regime, conditioning on survival means conditioning on a population of exactly two, and the quasi-stationary mean is 2. In the continuous-time model a survivor is a single individual that simply has not yet died, so the quasi-stationary mean is 1. The reciprocals are $\frac{1}{2}$ and 1, and there is the discrepancy.

At $p \rightarrow \frac{1}{2}^-$ the factor disappears, because both constants are governed by the same critical scaling. Expanding [\(28\)](#) near $p = \frac{1}{2}$ gives $A_c(p) = (1-2p)/(1-p) \rightarrow 2(1-2p)$, which is precisely [\(37\)](#). Near criticality the discreteness of the generational clock stops mattering; what survives is the universal 2ε behaviour, and the two models agree to leading order.

So the discrete curve is squeezed between $\frac{1}{2}A_c$ and its own critical asymptote, hugging the former at one end and the latter at the other. That is the whole content of [figure 9](#).

5.5 Searching for a closed form

Before the connection of [section 5.6](#) was appreciated, a number of methods were tried in the hope of an elementary formula for $A(p)$. Two are worth recording, partly for the negative result and partly because the way they failed was itself informative.

The first was the continuum route already described in [section 3.4](#): treat the map as an ODE in the large-generation limit and solve it. This gives the shape of the decay but, as noted there, leaves the constant of integration undetermined, which is to say it leaves $A(p)$ undetermined.

The second was symbolic regression by genetic algorithm. The concept is simple: generate fifty candidate functions at random, measure how closely each reproduces the target curve, retain the best five, and build a fresh population of fifty by varying those five. Measure again, and repeat the cycle of variation and selection many times over [\[6, 7\]](#).

The first implementation searched a space of polynomials, partly on a hunch that the answer might take that form and partly because such functions are simple to encode. In the minimal case a genome is just a coefficient list $[a_0, \dots, a_k]$ for

$$a_0 + a_1p + a_2p^2 + \dots + a_kp^k$$

up to some chosen maximum order k , with the initial population drawn uniformly at random. It quickly became apparent that many substantially different formulae produce curves visually indistinguishable from the target — but we wanted the exact one. On the supposition that a true formula might have rational coefficients, a second version searched

$$\frac{a_0}{b_0} + \frac{a_1}{b_1}p + \cdots + \frac{a_k}{b_k}p^k, \quad [a_i], [b_i] \in \mathbb{Z}^{k+1},$$

and a third extended the space to ratios of two such polynomials, with genome in $\mathbb{Z}^{4(k+1)}$. After numerous searches the best that emerged merely fitted the high-precision numerical data well:

$$\hat{A}_1(p) = \frac{\frac{1}{2}(1-2p)(1+2p)}{1 + \frac{1}{2}p - \frac{5}{2}p^2 - \frac{6}{5}p^3 + \frac{6}{5}p^4}. \quad (39)$$

This passes through $(0, \frac{1}{2})$ and $(\frac{1}{2}, 0)$ and looks convincing to the eye, but it fails the diagnostic of (37): differentiating at $p = \frac{1}{2}$ gives a gradient of $-2/0.55 = -3.636$ rather than the required -4 .

Two years after this work was set aside it was revisited with more capable tooling, using a symbolic tree search over progressively nested elementary functions of all kinds. The conclusion was the same: many functions of widely varying form resemble the target closely, and no satisfactory simple formula emerges even when solutions with fewer components are explicitly favoured. Two arbitrary picks from the sea of similarly performing candidates:

$$\begin{aligned} \hat{A}_2(p) = & 0.31696448624690005 \ln \left(\left| \cos \left(\left(\frac{1}{3.19801133824532} \right)^{1/4} \right. \right. \right. \\ & \left. \left. \left. + p \left[\sin \left(\cos(|p| (0.621865696665606 - p) \cos(e^p)) \right) + |p| \right] \right) \right| \right) + 0.5982282661889831 \end{aligned} \quad (40)$$

and, more concisely,

$$\hat{A}_3(p) = \frac{-0.616647881739505 e^{-0.552605335919693} e^p}{\sin(\cos p) - |p|} + 0.921964323679771. \quad (41)$$

(40) is quoted after removing redundancies such as a factor p/p that the program had introduced — neutral mutations, in effect, carrying no fitness cost and so never selected against.

The recurring defect is the same in every case. The candidates either fail to pass through $(\frac{1}{2}, 0)$ or fail to arrive there with gradient -4 , and near criticality this matters enormously: as $A \rightarrow 0$, an error in A of vanishing absolute size produces an $O(1)$ — eventually unbounded — error in the quasi-stationary mean $1/A$. A formula can look perfect on a plot of A and be useless on a plot of $1/A$.

A pragmatic patch is to splice the asymptotic onto a fitted curve, as in

$$\hat{A}_4(p) = \begin{cases} -0.04011483983619545 e^{e^{2p}} + 0.6079994336528085, & p < 0.499962, \\ 2 - 4p, & p \geq 0.499962, \end{cases} \quad (42)$$

which does work. But if one is going to splice, it is better to splice onto something exact, and [section 5.7](#) does so.

The deeper lesson from these searches is worth stating plainly, because it points directly at the next subsection. That an enormous number of very different formulae can behave almost identically over a bounded interval is not merely an inconvenience of the method; it is a symptom. Curve-fitting cannot distinguish a function that has a closed form from one that does not.

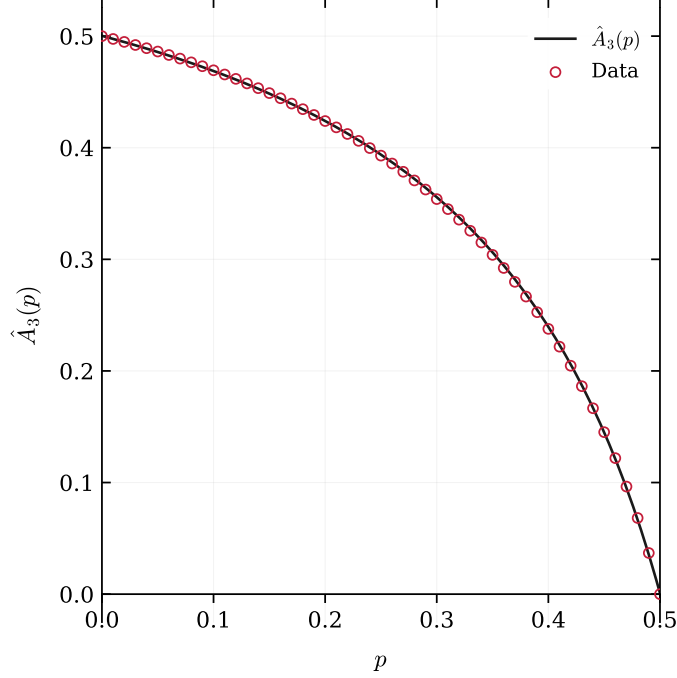


Figure 11: The symbolic-regression candidate $\hat{A}_3(p)$ of (41) (solid) against numerically computed values of $A(p)$ (red circles). The agreement is excellent to the eye across the whole interval, which is exactly the problem: the fit is not evidence of correctness. $\hat{A}_3$ misses the critical point by $\hat{A}_3(\frac{1}{2}) = 9.1 \times 10^{-4}$ rather than 0, and arrives there with gradient -3.63 instead of -4 . Errors of this size are invisible here but are fatal for $1/A$ near criticality.

5.6 The Koenigs connection

It was only some time after the search had been abandoned that a dynamical argument emerged explaining why no elementary closed form should be expected. In short, $A(p)$ is a value of the Koenigs linearising function of the logistic map, and that function is known to be non-elementary throughout the parameter range we need.

Definition 5.5 (Koenigs function). Let $f(z) = rz(1 - z)$ have a fixed point at $z = 0$ with multiplier $r = f'(0)$, and suppose $|r| \notin \{0, 1\}$. Koenigs [8] proved that there exists a unique function ψ_r , analytic near the origin, satisfying the Schröder equation

$$\psi_r(f(z)) = r \psi_r(z), \quad \psi_r(0) = 0, \quad \psi'_r(0) = 1. \quad (43)$$

The Koenigs function is a change of coordinate that straightens the dynamics: in the ψ coordinate the nonlinear map f becomes multiplication by r (figure 12). In our case $f(z) = rz(1 - z)$ is exactly (30), with $z = w_n$ and $f(z) = w_{n+1}$, and the multiplier is $r = 2p \in (0, 1)$ — an attracting fixed point, which is the classical Koenigs setting.

Proposition 5.6. For $p \in (0, \frac{1}{2})$ and $r = 2p$,

$$A(p) = 2 \psi_r\left(\frac{1}{2}\right). \quad (44)$$

Proof. Iterating (43) along the orbit $w_0, w_1, w_2, \dots$ of (30) gives $\psi(w_1) = r\psi(w_0)$, hence $\psi(w_0) =$

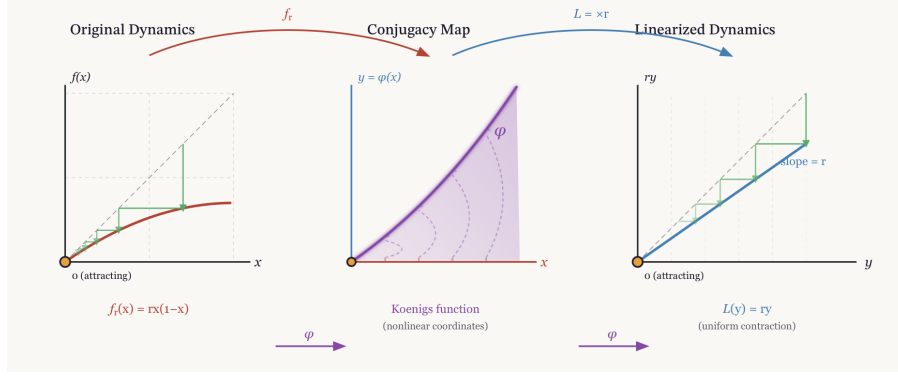


Figure 12: Koenigs linearisation of the logistic map near an attracting fixed point. The nonlinear dynamics $x \mapsto f_r(x) = rx(1-x)$ are conjugated by ψ_r to the linear map $y \mapsto ry$; the diagram displays the functional relation $\psi_r(f_r(x)) = r\psi_r(x)$ and the role of ψ_r as the coordinate change that straightens the dynamics.

$\psi(w_1)/r$; similarly $\psi(w_1) = \psi(w_2)/r$, so $\psi(w_0) = \psi(w_2)/r^2$; and by induction

$$\psi(w_0) = \frac{\psi(w_n)}{r^n} \quad \text{for every } n. \quad (45)$$

Since (45) holds for all n we may pass to the limit,

$$\psi(w_0) = \lim_{n \rightarrow \infty} \frac{\psi(w_n)}{r^n}.$$

Now the normalisation $\psi(0) = 0$, $\psi'(0) = 1$ means $\psi(z)/z \rightarrow 1$ as $z \rightarrow 0$, and $w_n = S_n/2 \rightarrow 0$ because $r < 1$. Replacing $\psi(w_n)$ by w_n in the limit is therefore legitimate, and

$$\psi(w_0) = \lim_{n \rightarrow \infty} \frac{w_n}{r^n}.$$

Comparing with (31) and using $w_0 = S_0/2 = \frac{1}{2}$ gives $A(p) = 2\psi_r(\frac{1}{2})$. $\square$

When is ψ_r elementary?

The answer is: essentially never. The logistic family is affinely conjugate to the canonical quadratic family $P_c(z) = z^2 + c$; substituting $x = -z/r + \frac{1}{2}$ in f_r gives P_c with

$$c = \frac{2r - r^2}{4}. \quad (46)$$

Under (46) the interval $r \in (0, 1)$ maps to $c \in (0, \frac{1}{4})$ (figure 13). The relevance of this is a rigidity theorem of Becker and Bergweiler [9]: for a rational map R of degree at least two with a fixed point of multiplier λ , $0 < |\lambda| < 1$, the associated Koenigs linearisation satisfies an algebraic differential equation only when R is analytically conjugate to a monomial map $z \mapsto z^{\pm d}$ or to a Chebyshev polynomial $\pm T_d$. For the quadratic family these exceptional cases are exactly $c = 0$ and $c = -2$, corresponding under (46) to $r = 2$ and $r = 4$ respectively.

Since every elementary function in the sense of Liouville — algebraic functions, exponentials, logarithms and their finite combinations — satisfies an algebraic differential equation, a function that satisfies none (a *hypertranscendental* function) cannot be elementary. And

$$(0, \frac{1}{4}) \cap \{0, -2\} = \emptyset,$$

so no r in our range is exceptional: for each fixed $r \in (0, 1)$, the Koenigs function ψ_r is hypertranscendental, and in particular not elementary.

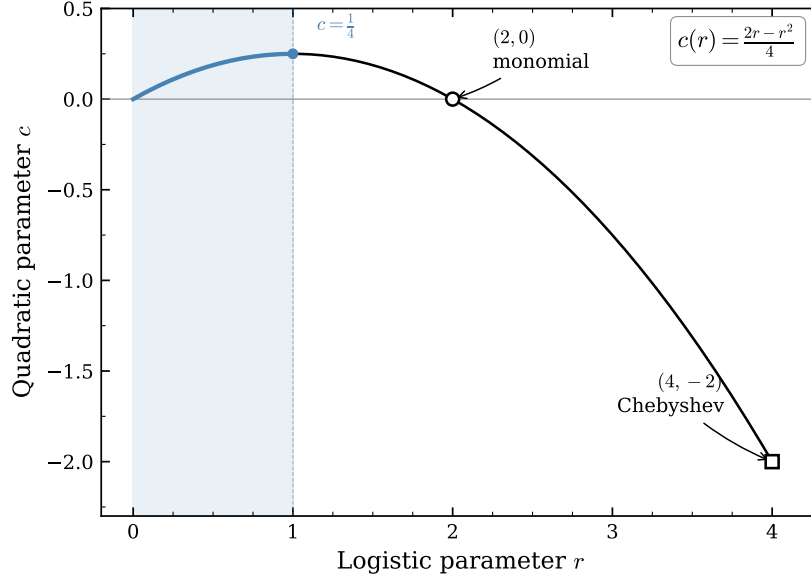


Figure 13: The affine parameter correspondence $c(r) = (2r - r^2)/4$ between the logistic family $f_r(x) = rx(1 - x)$ and the quadratic family $P_c(z) = z^2 + c$. The shaded segment is $r \in (0, 1)$, the subcritical branching window, which maps to $c \in (0, \frac{1}{4})$. The two integrable parameters, $r = 2$ ($c = 0$, monomial) and $r = 4$ ($c = -2$, Chebyshev), lie outside it.

The two exceptional parameters are worth writing out, both as a check on the machinery and because they show just how special the solvable cases are.

The monomial case, $r = 2$. Here $f_2(x) = 2x(1 - x)$. The substitution $u = 1 - 2x$ gives

$$u_{n+1} = 1 - 2(2x_n(1 - x_n)) = 1 - 4x_n + 4x_n^2 = (1 - 2x_n)^2 = u_n^2,$$

and the map $u \mapsto u^2$ is linearised by the logarithm, since $\ln(u^2) = 2 \ln u$. Returning to x and imposing the normalisation of [definition 5.5](#),

$$\psi_2(x) = -\frac{1}{2} \ln(1 - 2x), \quad (47)$$

which is readily verified: $\psi_2(f_2(x)) = -\frac{1}{2} \ln((1 - 2x)^2) = 2[-\frac{1}{2} \ln(1 - 2x)] = 2\psi_2(x)$.

The Chebyshev case, $r = 4$. Here $f_4(x) = 4x(1 - x)$ is conjugate to $T_2(z) = 2z^2 - 1$ by a trigonometric substitution. Putting $x = \sin^2 \theta$,

$$f_4(x) = 4 \sin^2 \theta \cos^2 \theta = (2 \sin \theta \cos \theta)^2 = \sin^2(2\theta),$$

so the map is a doubling $\theta \mapsto 2\theta$ in the θ coordinate. The multiplier at the origin is 4, not 2, which suggests that the linearisation involves θ^2 ; and indeed

$$\psi_4(x) = (\arcsin \sqrt{x})^2. \quad (48)$$

Using $\arcsin(2u\sqrt{1-u^2}) = 2 \arcsin u$ with $u = \sqrt{x}$, $\psi_4(f_4(x)) = (2 \arcsin \sqrt{x})^2 = 4\psi_4(x)$; and as $x \rightarrow 0$, $\arcsin \sqrt{x} \approx \sqrt{x}$, so $\psi_4(x) \approx x$ and $\psi_4'(0) = 1$.

Remark 5.7 (Attracting versus repelling). At $r = 2$ and $r = 4$ the multiplier at the origin is $|r| > 1$, so the origin is *repelling*. The functions (47) and (48) are therefore Poincaré-type linearisations rather than attracting-Koenigs maps, and neither parameter lies in the branching window $r = 2p \in [0, 1]$ in any case. They are recorded here to show what a solvable case looks like, not because they bear on $A(p)$.

What this does and does not establish

It is important to be precise about the status of the conclusion, because it is easy to overstate.

What is established is that for each *fixed* $r \in (0, 1)$ the function $z \mapsto \psi_r(z)$ is hypertranscendental. Combined with [proposition 5.6](#), this says that $A(p)$ is a value of a non-elementary function, evaluated at the fixed point $z = \frac{1}{2}$, for every p in the subcritical range. It is a strong indication that the search of [section 5.5](#) was doomed, and it explains the phenomenology of that search: the object being fitted is defined only by the dynamics that generate it.

What is *not* established is that the map $p \mapsto A(p)$ admits no elementary description. Hypertranscendence of ψ_r in its argument z is a statement about one variable; $A(p) = 2\psi_{2p}(\frac{1}{2})$ varies the parameter and freezes the argument, and the two questions are logically distinct. A rigorous negative result for $p \mapsto A(p)$ would require a transcendence statement about the values, not the functions, and none is claimed here. The honest summary is that no closed form is known, that the structural reason for expecting none is the one just given, and that the numerical evidence of [section 5.5](#) is entirely consistent with there being none.

It is worth seeing where in the quadratic family our parameters sit, because the picture makes the difficulty look structural rather than accidental. The main cardioid of the Mandelbrot set — the set of c for which $z^2 + c$ has an attracting fixed point — is parametrised by that point’s multiplier λ through $c = \lambda/2 - \lambda^2/4$, which is exactly (46) with $\lambda = r$. The branching window $r \in (0, 1)$ is therefore the real diameter of the main cardioid, traversed from the centre $c = 0$ out to the parabolic cusp $c = \frac{1}{4}$, where $\lambda = 1$ and the process turns critical. [Figure 14](#) places the segment in context; the two integrable parameters $c = 0$ and $c = -2$ are marked, and the open window contains neither.

Throughout the window the Julia set is a quasicircle bounding the basin of the attracting fixed point,¹ and it is a genuine fractal for every c in the interval. The Koenigs function is the conformal coordinate that linearises the dynamics inside that basin, so its natural domain is bounded by exactly this curve. A function whose domain of definition has no smooth description is not a promising candidate for a closed form, and $A(p)$, being one of its values, inherits the difficulty. The contrast with the integrable cases is sharp: at $c = 0$ the Julia set degenerates to the unit circle, and it is precisely there that the linearisation becomes elementary, (47).

5.7 Computing $A(p)$ in practice

In the absence of an exact formula we are left with approximation, and the two regimes of [section 5.2](#) suggest how to split the work.

Over most of the interval, direct iteration is entirely satisfactory: the infinite product (32) converges quickly, its partial products bound $A(p)$ from above at every order, and modest numbers of iterations give many digits. The difficulty is confined to a neighbourhood of $p = \frac{1}{2}$, and it has two parts. First, as $A(p) \rightarrow 0$ the quantity of interest $1/A$ diverges, so a fixed absolute error in

¹An interactive visualiser accompanying this chapter plots ψ_r over the complex plane by the product form $\psi_r(z) = z \prod_{n \geq 0} (1 - f_r^{(n)}(z))$ — the same telescoping identity used in [section 5.1](#) — colouring by the phase of ψ_r so that the linearised coordinate grid and the fractal basin boundary are both visible ([Koenigs_plotter/gem_p1.html](#)).

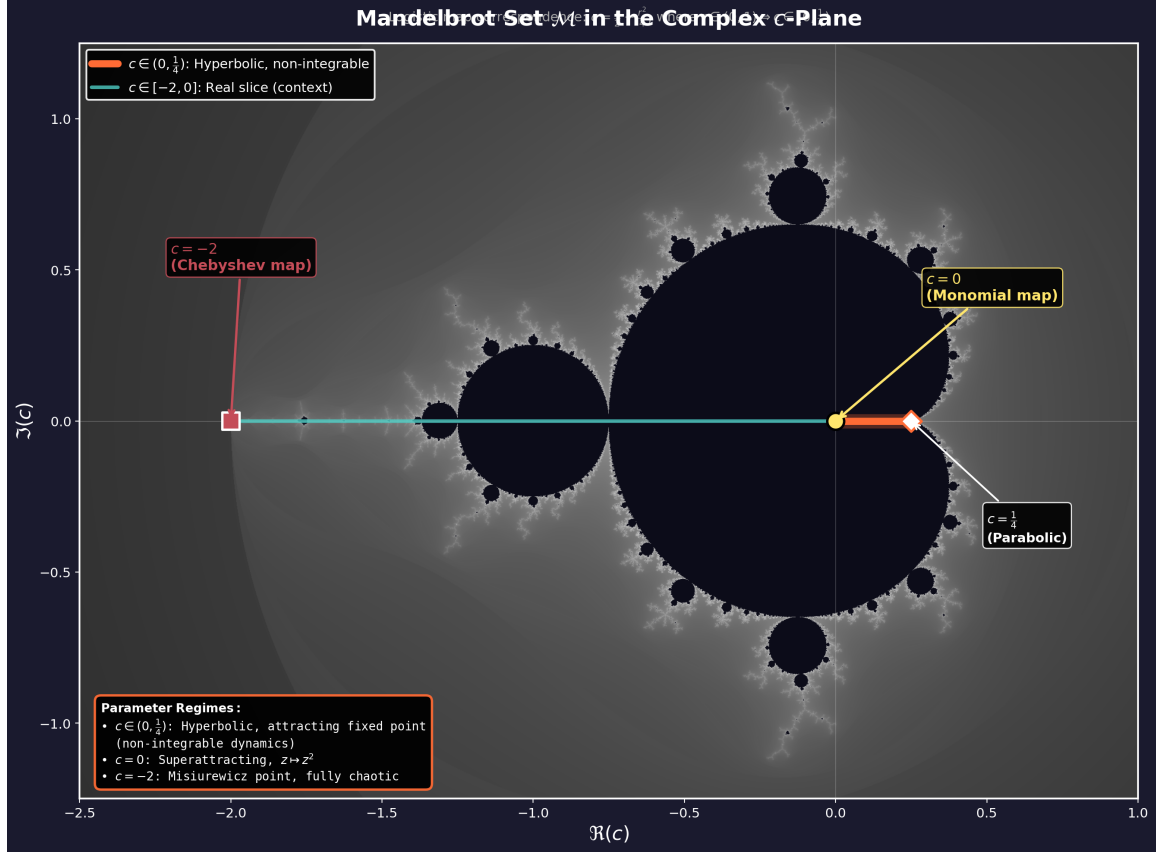


Figure 14: The Mandelbrot set in the complex c -plane, with the interval $c \in (0, \frac{1}{4})$ highlighted. Under the correspondence $c = (2r - r^2)/4$ of (46) this is the image of the subcritical branching window $r = 2p \in (0, 1)$, and it forms the real diameter of the main cardioid, running from the centre $c = 0$ to the parabolic cusp $c = \frac{1}{4}$ — which is criticality. The two integrable parameters are marked: $c = 0$ (monomial, reached in logistic coordinates at $r = 2$) and $c = -2$ (Chebyshev, $r = 4$), the latter at the far tip of the real slice. Note that $c(r) = c(2 - r)$, so each $c \in (0, \frac{1}{4})$ is the image of two logistic parameters, r and $2 - r$; these correspond to the two fixed points of $z^2 + c$, with multipliers r and $2 - r$, and it is the attracting one, $r < 1$, that governs $A(p)$.

A becomes an unbounded error in the quasi-stationary mean. Second — and this is the binding constraint — as the process is defined ever closer to criticality its lifetime, though still finite, grows enormously, and S_n recedes to zero ever more slowly. The iteration must be carried to a generation at which S_n is essentially unchanging, and table 1 shows what that costs: about 10^2 iterations at $p = 0.2$, 10^3 at $p = 0.45$, but nearly 5×10^5 at $p = 0.4999$.

The remedy is to hand the near-critical region to the asymptotic (37), which is exact in the limit and cheap everywhere. This gives

$$A(p) \approx \begin{cases} \frac{1}{2} \prod_{k=1}^N \left(1 - \frac{S_k}{2}\right), & p < c, \quad N \gg 1, \\ 2 - 4p, & p \geq c, \end{cases} \quad (49)$$

with the crossover c chosen just below $\frac{1}{2}$. Remark 5.4 indicates where to put it: the relative error of the asymptotic is about $\varepsilon \ln(1/\varepsilon)$, so at $c = 0.4999$ ($\varepsilon = 2 \times 10^{-4}$) the asymptotic is already accurate

to about two parts in 10^3 , while the product still converges in a few hundred thousand iterations. Any c in the range 0.499–0.4999 gives a scheme accurate to better than one per cent throughout, and the two branches can be blended over a narrow window if continuity of the derivative matters.

This is the recipe used for the numerical values quoted throughout this chapter, including those in [figure 5](#) and [table 1](#).

6 Absorption models and the method of characteristics

In previous work we have been able to describe the birth–death dynamics of pathogenic particles inside a host cell that can rupture. We now wish to extend that understanding to cases in which several host cells share a medium, and this requires deciding what happens when the first of them ruptures. The shared medium suddenly gains a potentially large cohort of pathogenic particles, available for subsequent absorption by the remaining hosts.

The analysis so far has assumed *immediate transfer*: the entire pathogenic contents of a rupturing cell relocate at once to the interior of another host. That is a strong assumption. A more realistic alternative is an *absorption period*: following a rupture, absorption events redistribute the released cohort among the remaining hosts, and only when every particle has been taken up do birth, death and rupture resume. Working towards such absorption models, we begin with the case in which a pathogenic cohort infiltrates a single host cell.

Three models are treated in increasing order of difficulty. In the first two the particles are conserved or only lost, the state space is finite, and an independence argument delivers the generating function without any need to solve a partial differential equation. In the third, replication inside the host opens an infinite state space and the independence shortcut fails, so the PDE must be confronted directly. The method of characteristics is developed on the second model, where the answer is already known, and then applied to the third.

6.1 Absorption only

The simplest absorption process has a single host cell that takes in pathogenic particles, with no subsequent dynamics — no birth and no death of pathogen within the host.

Introduce $\mathbf{X}_t$ and $\mathbf{Y}_t$ for the number of pathogenic particles inside and outside the host cell respectively; the same labelling is used for every version of the single-host model below. Events are assumed stochastic, with exponentially distributed inter-event times, a property that follows from the assumption that event rates do not depend on time. The continuous-time stochastic process $\{(\mathbf{X}_t, \mathbf{Y}_t) : t \in [0, \infty)\}$ therefore has the memoryless Markov property.

With only absorption to consider, the model is specified by the following outcome-likelihoods over an infinitesimal duration Δt , taken in the limit to zero:

$$\begin{aligned} \Pr\{\Delta\mathbf{X} = 1, \Delta\mathbf{Y} = -1 \mid \mathbf{Y}_t = j\} &= j\alpha\Delta t, & (\text{absorption}) \\ \Pr\{\Delta\mathbf{X} = 0, \Delta\mathbf{Y} = 0 \mid \mathbf{Y}_t = j\} &= 1 - j\alpha\Delta t, & (\text{no event}) \end{aligned}$$

where $\Delta\mathbf{X} = \mathbf{X}_{t+\Delta t} - \mathbf{X}_t$, and similarly for $\mathbf{Y}$. Note the factor j : each of the extracellular particles is independently subject to the same probability $\alpha\Delta t$ of being taken up. The initial condition is $\mathbf{X}_0 = 0, \mathbf{Y}_0 = N$ — a starter population of census N occupying the extracellular medium.

Defining state probabilities $p_{i,j}(t) = \Pr\{\mathbf{X}_t = i, \mathbf{Y}_t = j\}$, the forward Kolmogorov equations are

$$\frac{dp_{i,j}}{dt} = \alpha(j+1)p_{i-1,j+1} - \alpha j p_{i,j}. \quad (50)$$

With the probability generating function

$$G(x, y, t) = \sum_i \sum_j p_{i,j}(t) x^i y^j \quad (= \mathbb{E}[x^{\mathbf{X}_t} y^{\mathbf{Y}_t}]), \quad (51)$$

we can convert (50) into a PDE. Multiply each equation by $x^i y^j$,

$$\frac{\partial p_{i,j} x^i y^j}{\partial t} = \alpha(j+1) p_{i-1,j+1} x^i y^j - \alpha j p_{i,j} x^i y^j,$$

and sum over i and j for non-zero $p_{i,j}(t)$:

$$\sum_{i=0}^N \sum_{j=0}^N \frac{\partial p_{i,j} x^i y^j}{\partial t} = \alpha \sum_{i=1}^{N+1} \sum_{j=-1}^{N-1} (j+1) p_{i-1,j+1} x^i y^j - \alpha \sum_{i=0}^N \sum_{j=0}^N j p_{i,j} x^i y^j. \quad (52)$$

(The sums are finite because $p_{i,j}(t) = 0$ for all t whenever $i+j \neq N$: pathogenic particles in this model are neither created nor destroyed.) Shifting indices in the first double sum and rearranging,

$$\sum_{i=0}^N \sum_{j=0}^N \frac{\partial p_{i,j} x^i y^j}{\partial t} = \alpha \left(x \sum_{i=0}^N \sum_{j=0}^N j p_{i,j} x^i y^{j-1} - y \sum_{i=0}^N \sum_{j=0}^N j p_{i,j} x^i y^{j-1} \right),$$

which is a statement about partial derivatives,

$$\sum_{i=0}^N \sum_{j=0}^N \frac{\partial p_{i,j} x^i y^j}{\partial t} = \alpha (x - y) \sum_{i=0}^N \sum_{j=0}^N \frac{\partial p_{i,j} x^i y^j}{\partial y}.$$

Recognising (51),

$$\frac{\partial G(x, y, t)}{\partial t} = \alpha (x - y) \frac{\partial G(x, y, t)}{\partial y}, \quad G(x, y, 0) = y^N. \quad (53)$$

This first-order linear initial value problem proves surprisingly awkward. The method of characteristics, though applicable in principle to equations in three variables, led at first only to mystification. As it happens, the solution can be obtained by other means, and it is worth doing so both because the answer is needed and because it provides a check on the characteristics machinery developed in [section 6.3](#).

Exploiting independence

The function G above was specific to $\mathbf{X}_0 = 0$, $\mathbf{Y}_0 = N$. Define more generally

$$G_n(x, y, t) = \sum_i \sum_j \Pr \{ \mathbf{X}_t = i, \mathbf{Y}_t = j \mid \mathbf{X}_0 = 0, \mathbf{Y}_0 = n \} x^i y^j.$$

In a process of pure absorption the N particles are taken up one by one, and if we follow a single particle its behaviour does not depend on the movements of its neighbours. Because of this independence, $\mathbf{X}_t$ and $\mathbf{Y}_t$ decompose into sums of n single-particle variables,

$$\begin{aligned} \mathbf{X}_t &= \mathbf{X}_t^{(1)} + \mathbf{X}_t^{(2)} + \dots + \mathbf{X}_t^{(n)}, \\ \mathbf{Y}_t &= \mathbf{Y}_t^{(1)} + \mathbf{Y}_t^{(2)} + \dots + \mathbf{Y}_t^{(n)}, \end{aligned}$$

from which

$$G_n(x, y, t) = (G_1(x, y, t))^n, \quad (54)$$

since

$$\begin{aligned} G_n(x, y, t) &= \mathbb{E} \left(x^{\sum_{k=1}^n \mathbf{X}_t^{(k)}} y^{\sum_{k=1}^n \mathbf{Y}_t^{(k)}} \right) = \mathbb{E} \left(\prod_{k=1}^n x^{\mathbf{X}_t^{(k)}} \prod_{k=1}^n y^{\mathbf{Y}_t^{(k)}} \right) \\ &= \prod_{k=1}^n \mathbb{E} \left(x^{\mathbf{X}_t^{(k)}} y^{\mathbf{Y}_t^{(k)}} \right) = (G_1(x, y, t))^n. \end{aligned}$$

Now consider a single initial particle, $\mathbf{X}_0 = 0$, $\mathbf{Y}_0 = 1$. It is eventually absorbed, and that is the final state of the system: no other states are reachable, so $p_{i,j}(t) = 0$ for all t whenever $i + j \neq 1$. The forward Kolmogorov equations reduce to

$$\frac{dp_{0,1}(t)}{dt} = -\alpha p_{0,1}(t), \quad \frac{dp_{1,0}(t)}{dt} = \alpha p_{0,1}(t),$$

giving

$$p_{0,1}(t) = e^{-\alpha t}, \quad p_{1,0}(t) = 1 - e^{-\alpha t}.$$

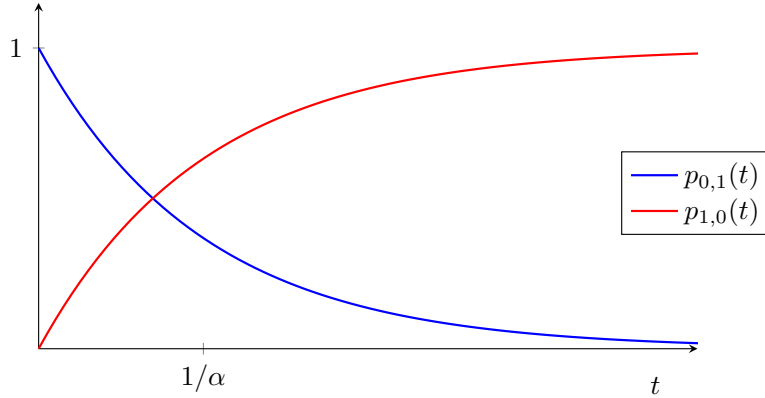


Figure 15: State probabilities for the single-particle absorption-only model. The particle begins outside the cell ($p_{0,1}$, decaying exponentially at rate α) and ends inside it ($p_{1,0}$). There is nowhere else for it to go, so the two curves sum to one at every time.

These being the only two non-zero state probabilities of the single-particle system,

$$G_1(x, y, t) = p_{1,0}(t) x + p_{0,1}(t) y = (1 - e^{-\alpha t}) x + e^{-\alpha t} y,$$

and applying (54) gives the generating function for any initial census,

$$G_n(x, y, t) = \left((1 - e^{-\alpha t}) x + e^{-\alpha t} y \right)^n. \quad (55)$$

This does satisfy (53) together with its initial condition $G_n(x, y, 0) = y^n$. The form is recognisable: $\mathbf{X}_t \sim \text{Binomial}(n, 1 - e^{-\alpha t})$, as it must be, since the n particles are absorbed independently and each has been taken up by time t with probability $1 - e^{-\alpha t}$.

6.2 Absorption–death

In reality a pathogenic particle resident in a host cell may die. Keeping $\mathbf{X}_t$ and $\mathbf{Y}_t$ for the intracellular and extracellular counts, death is incorporated as follows:

$$\begin{aligned} \Pr\{\Delta\mathbf{X} = 1, \Delta\mathbf{Y} = -1 \mid \mathbf{Y}_t = j\} &= j\alpha\Delta t, & (\text{absorption}) \\ \Pr\{\Delta\mathbf{X} = -1, \Delta\mathbf{Y} = 0 \mid \mathbf{X}_t = i\} &= i\mu\Delta t, & (\text{death}) \\ \Pr\{\Delta\mathbf{X} = 0, \Delta\mathbf{Y} = 0 \mid \mathbf{Y}_t = j, \mathbf{X}_t = i\} &= 1 - (j\alpha + i\mu)\Delta t, & (\text{no event}) \end{aligned}$$

with forward Kolmogorov equations

$$\frac{dp_{i,j}}{dt} = \alpha(j+1)p_{i-1,j+1} + \mu(i+1)p_{i+1,j} - (\alpha j + \mu i)p_{i,j}. \quad (56)$$

An analogous derivation to that of [section 6.1](#) leads to the first-order linear PDE

$$\frac{\partial G(x, y, t)}{\partial t} = \mu(1-x)\frac{\partial G(x, y, t)}{\partial x} + \alpha(x-y)\frac{\partial G(x, y, t)}{\partial y}, \quad G(x, y, 0) = y^N, \quad (57)$$

and the same familiar difficulty. But once again the solution can be constructed via [\(54\)](#), since individual particles remain independent in their movements.

Without birth events the number of particles cannot exceed its initial value, so for an isolated particle the generating function has just three terms,

$$G_1(x, y, t) = p_{0,0}(t) + p_{1,0}(t)x + p_{0,1}(t)y.$$

The state probabilities follow from

$$\frac{dp_{0,1}(t)}{dt} = -\alpha p_{0,1}(t), \quad \frac{dp_{1,0}(t)}{dt} = \alpha p_{0,1}(t) - \mu p_{1,0}(t),$$

whence

$$p_{0,1}(t) = e^{-\alpha t}, \quad p_{1,0}(t) = \frac{\alpha}{\mu - \alpha} (e^{-\alpha t} - e^{-\mu t}), \quad p_{0,0}(t) = 1 - p_{0,1}(t) - p_{1,0}(t). \quad (58)$$

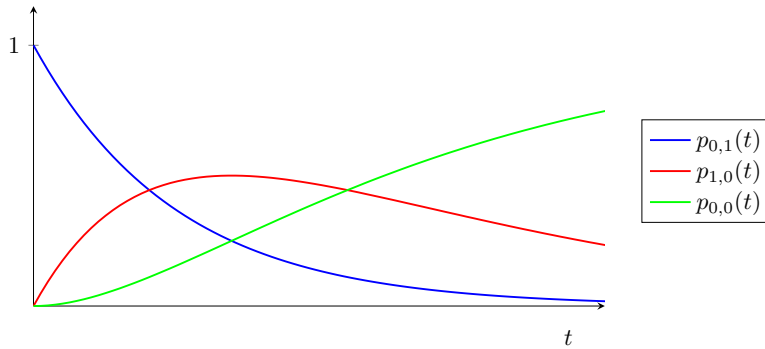


Figure 16: State probabilities for the single-particle absorption–death model, [\(58\)](#). The particle starts outside ($p_{0,1}$, blue), is absorbed into the cell ($p_{1,0}$, red, which rises and then decays as death takes over), and finally dies ($p_{0,0}$, green, absorbing).

Applying [\(54\)](#) gives the general generating function

$$G_n(x, y, t) = \left(1 - \frac{\mu}{\mu - \alpha} e^{-\alpha t} + \frac{\alpha}{\mu - \alpha} e^{-\mu t} + \frac{\alpha}{\mu - \alpha} (e^{-\alpha t} - e^{-\mu t})x + e^{-\alpha t}y\right)^n. \quad (59)$$

6.3 The method of characteristics, tested on absorption–death

Before continuing to absorption–birth–death, where replication may occur within the host cell and the independence shortcut is unavailable, it is worth revisiting absorption–death from the point of view of its PDE. Having (59) already in hand makes this an ideal test case.²

Introduce a change of variables $(x, y, t) \rightarrow (s, r, \tau)$,

$$G(x, y, t) = G(x(s, r, \tau), y(s, r, \tau), t(s, r, \tau)) \equiv G(s, r, \tau).$$

The method works by seeking solutions along the characteristic curves of constant s and r , on which G is a function of τ alone. To fix the parametrisation we impose an initial surface,

$$x(s, r, 0) = s, \quad y(s, r, 0) = r, \quad t(s, r, 0) = 0. \quad (60)$$

With G a function of τ only along a characteristic, the chain rule gives

$$\frac{dG}{d\tau} = \frac{\partial G}{\partial t} \frac{dt}{d\tau} + \frac{\partial G}{\partial x} \frac{dx}{d\tau} + \frac{\partial G}{\partial y} \frac{dy}{d\tau},$$

which is to be compared with (57) written as

$$0 = -\frac{\partial G}{\partial t} + \mu(1-x)\frac{\partial G}{\partial x} + \alpha(x-y)\frac{\partial G}{\partial y}.$$

Comparing coefficients yields the characteristic system

$$\frac{dG}{d\tau} = 0, \quad (61)$$

$$\frac{dt}{d\tau} = -1, \quad (62)$$

$$\frac{dx}{d\tau} = \mu(1-x), \quad (63)$$

$$\frac{dy}{d\tau} = \alpha(x-y). \quad (64)$$

The equation for G , (61). G is constant along characteristics, $G = \phi(s, r)$. Under (60) the initial condition $G(x, y, 0) = y^N$ becomes $G(s, r, 0) = r^N$, so $\phi(s, r) = r^N$ and

$$G(s, r, \tau) = r^N. \quad (65)$$

The whole problem thus reduces to expressing r in terms of (x, y, t) .

The equation for t , (62). $t = -\tau + c(s, r)$, and (60) fixes $c = 0$:

$$t = -\tau. \quad (66)$$

The characteristic parameter runs backwards in time, which is what allows the initial surface to be reached from any (x, y, t) .

²I am grateful to Benjamin Sharp for the advice on how to apply the method of characteristics to three-variate problems that unlocked this section.

The equation for x , (63). Separating variables gives $1 - x = c(s, r)e^{-\mu\tau}$, and (60) gives $c = 1 - s$:

$$1 - x = (1 - s)e^{-\mu\tau}. \quad (67)$$

The equation for y , (64). Substituting (67),

$$\frac{dy}{d\tau} = \alpha(1 - (1 - s)e^{-\mu\tau} - y),$$

and solving with integrating factor $e^{\alpha\tau}$,

$$y = 1 - \frac{\alpha}{\alpha - \mu}(1 - s)e^{-\mu\tau} + c(s, r)e^{-\alpha\tau}.$$

Applying (60), $c(s, r) = r + \frac{\alpha}{\alpha - \mu}(1 - s) - 1$, so

$$y = 1 - \frac{\alpha}{\alpha - \mu}(1 - s)e^{-\mu\tau} + \left(r + \frac{\alpha}{\alpha - \mu}(1 - s) - 1\right)e^{-\alpha\tau}. \quad (68)$$

Recovering the solution. Rearranging (68) for r , substituting s from (67) and $\tau = -t$ from (66), gives

$$r = 1 - \frac{\mu}{\mu - \alpha}e^{-\alpha t} + \frac{\alpha}{\mu - \alpha}e^{-\mu t} + \frac{\alpha}{\mu - \alpha}(e^{-\alpha t} - e^{-\mu t})x + e^{-\alpha t}y,$$

and then (65) reproduces (59) exactly. The two routes agree, and the machinery is ready for the case where only one of them is available.

6.4 Absorption–birth–death

Introducing replication for particles inside the host cell, we shall see that although the independence property is still present, it no longer serves as a shortcut. Given a single initial infectious particle, birth events make infinitely many states reachable, so $G_1(x, y, t)$ is an infinite sum and there is nothing to be gained by factorising. Fortunately we are now equipped to solve the PDE directly, and that is the approach taken.

With $\mathbf{X}_t$ and $\mathbf{Y}_t$ as before, birth is added to the transition probabilities:

$$\begin{aligned} \Pr\{\Delta\mathbf{X} = 1, \Delta\mathbf{Y} = -1 \mid \mathbf{Y}_t = j\} &= j\alpha\Delta t, & \text{(absorption)} \\ \Pr\{\Delta\mathbf{X} = 1, \Delta\mathbf{Y} = 0 \mid \mathbf{X}_t = i\} &= i\beta\Delta t, & \text{(birth)} \\ \Pr\{\Delta\mathbf{X} = -1, \Delta\mathbf{Y} = 0 \mid \mathbf{X}_t = i\} &= i\mu\Delta t, & \text{(death)} \\ \Pr\{\Delta\mathbf{X} = 0, \Delta\mathbf{Y} = 0 \mid \mathbf{Y}_t = j, \mathbf{X}_t = i\} &= 1 - (j\alpha + i\mu + i\beta)\Delta t, & \text{(no event)} \end{aligned}$$

with forward Kolmogorov equation

$$\frac{\partial p_{i,j}}{\partial t} = \alpha(j+1)p_{i-1,j+1} + \mu(i+1)p_{i+1,j} + \beta(i-1)p_{i-1,j} - (\alpha j + \mu i + \beta i)p_{i,j}. \quad (69)$$

The same reasoning as before gives the generating-function PDE

$$\frac{\partial G}{\partial t} = \left(\mu - (\mu + \beta)x + \beta x^2\right) \frac{\partial G}{\partial x} + \alpha(x - y) \frac{\partial G}{\partial y}, \quad G(x, y, 0) = y^N. \quad (70)$$

Reparametrising $(x, y, t) \rightarrow (s, r, \tau)$ as in section 6.3 gives the characteristic system

$$\frac{dG}{d\tau} = 0, \quad (71)$$

$$\frac{dt}{d\tau} = -1, \quad (72)$$

$$\frac{dx}{d\tau} = \mu - (\mu + \beta)x + \beta x^2, \quad (73)$$

$$\frac{dy}{d\tau} = \alpha(x - y), \quad (74)$$

with the same initial conditions (60).

The equations for G and t , (71)–(72). As before,

$$G(s, r, \tau) = r^N, \quad t = -\tau. \quad (75)$$

The equation for x , (73). This is a Riccati equation of the same shape as the one governing the survival probability of section 3.4. Its right-hand side factorises, and it is worth noting that the roots are elementary rather than the surd expression one gets from mechanically applying the quadratic formula:

$$\mu - (\mu + \beta)x + \beta x^2 = \beta(x - x_+)(x - x_-), \quad x_+ = 1, \quad x_- = \frac{\mu}{\beta}. \quad (76)$$

The discriminant is $(\frac{\mu+\beta}{2\beta})^2 - \frac{\mu}{\beta} = (\frac{\beta-\mu}{2\beta})^2$, a perfect square, so the surds cancel. It is convenient to name the combination that appears throughout,

$$b_1 = \beta(x_+ - x_-) = \beta - \mu, \quad (77)$$

the net intracellular growth rate. Separating variables in (73) and applying (60) gives the characteristic in the compact form

$$\frac{x - x_+}{x - x_-} = \left(\frac{s - x_+}{s - x_-} \right) e^{b_1 \tau}, \quad (78)$$

or, solved for x ,

$$x(\tau) = \frac{x_+(s - x_-) - x_-(s - x_+)e^{b_1 \tau}}{(s - x_-) - (s - x_+)e^{b_1 \tau}}. \quad (79)$$

The equation for y , (74). The last characteristic equation is $dy/d\tau = \alpha(x - y)$, and with an integrating factor it becomes

$$\frac{d}{d\tau} (e^{\alpha\tau} y) = \alpha e^{\alpha\tau} x(\tau). \quad (80)$$

The integral on the right is where the difficulty lies. Rather than substitute (79) and split the resulting quotient — which leads to two hypergeometric integrals and a good deal of bookkeeping — it is cleaner to expand $x(\tau)$ first. Writing $x - x_- = (x - x_+) + (x_+ - x_-)$ in (78) and solving for x gives

$$x(\tau) = x_+ + (x_+ - x_-) \frac{\zeta(\tau)}{1 - \zeta(\tau)}, \quad \zeta(\tau) = \left(\frac{s - x_+}{s - x_-} \right) e^{b_1 \tau}, \quad (81)$$

so that, for $|\zeta| < 1$,

$$x(\tau) = x_+ + (x_+ - x_-) \sum_{n=1}^{\infty} \zeta(\tau)^n.$$

Every term is now a pure exponential in τ and integrates immediately. Writing $Z = (s - x_+)/ (s - x_-)$, so that $\zeta(\tau) = Ze^{b_1\tau}$ and $Z^n e^{nb_1\tau} = \zeta^n$,

$$\begin{aligned} \alpha \int e^{\alpha\tau} x(\tau) d\tau &= \alpha x_+ \int e^{\alpha\tau} d\tau + \alpha(x_+ - x_-) \sum_{n=1}^{\infty} Z^n \int e^{(\alpha+nb_1)\tau} d\tau \\ &= e^{\alpha\tau} \left[x_+ + \alpha(x_+ - x_-) \sum_{n=1}^{\infty} \frac{\zeta^n}{\alpha + nb_1} \right] + \text{const.} \end{aligned}$$

Setting $\delta = \alpha/b_1$ and using $\alpha(x_+ - x_-)/b_1 = \alpha/\beta$ by (77), this defines the function that carries the whole solution:

$$\Psi(\zeta) = x_+ + \frac{\alpha}{\beta} \sum_{n=1}^{\infty} \frac{\zeta^n}{n + \delta}, \quad \delta = \frac{\alpha}{b_1} = \frac{\alpha}{\beta - \mu}. \quad (82)$$

So $\alpha \int e^{\alpha\tau} x(\tau) d\tau = e^{\alpha\tau} \Psi(\zeta(\tau)) + \text{const}$, and integrating (80),

$$e^{\alpha\tau} y = e^{\alpha\tau} \Psi(\zeta(\tau)) + \text{const.}$$

The initial condition $y(s, r, 0) = r$ with $\zeta(0) = Z$ fixes the constant as $r - \Psi(Z)$, so

$$r = e^{\alpha\tau} [y - \Psi(\zeta(\tau))] + \Psi(Z). \quad (83)$$

Finally, put $\tau = -t$ and use (78): writing

$$W = \frac{x - x_+}{x - x_-}$$

we have $\zeta(\tau) = W$ and $Z = We^{-b_1\tau} = We^{b_1t}$. Substituting into (83) and applying (75) gives the solution.

Proposition 6.1 (Generating function for absorption–birth–death). *With $x_+ = 1$, $x_- = \mu/\beta$, $b_1 = \beta - \mu$, $\delta = \alpha/b_1$, $W = (x - x_+)/ (x - x_-)$ and Ψ as in (82),*

$$\boxed{G(x, y, t) = \left\{ e^{-\alpha t} [y - \Psi(W)] + \Psi(We^{b_1t}) \right\}^N.} \quad (84)$$

Three checks confirm the result. At $t = 0$ the two Ψ terms cancel and $G = y^N$, the required initial condition. At $x = y = 1$ we have $W = 0$ and $\Psi(0) = x_+ = 1$, so $G = 1$ and probability is conserved. And setting $x = 1$ alone gives

$$G(1, y, t) = (e^{-\alpha t} y + 1 - e^{-\alpha t})^N,$$

so $\mathbf{Y}_t \sim \text{Binomial}(N, e^{-\alpha t})$ — exactly right, since the extracellular particles are absorbed independently at rate α and what happens inside the cell cannot affect them. Beyond these, (84) has been verified numerically against direct integration of the master equation (69), agreeing to around 10^{-13} across sub- and supercritical intracellular regimes and for $N = 1, 2, 3$.

Hypergeometric form

The series in (82) is a Gauss hypergeometric function in disguise. With

$${}_2F_1(a, b; c; z) = \sum_{n=0}^{\infty} \frac{(a)_n (b)_n}{(c)_n} \frac{z^n}{n!}, \quad (q)_n = \begin{cases} 1, & n = 0, \\ q(q+1) \cdots (q+n-1), & n \geq 1, \end{cases} \quad (85)$$

$(q)_n$ being the rising Pochhammer symbol, the case $a = 1$, $b = \delta$, $c = \delta + 1$ collapses to a single ratio. Since $(1)_n = n!$ and

$$\frac{(\delta)_n}{(\delta + 1)_n} = \frac{\delta(\delta+1) \cdots (\delta+n-1)}{(\delta+1) \cdots (\delta+n-1)(\delta+n)} = \frac{\delta}{\delta+n}, \quad (86)$$

we have

$${}_2F_1(1, \delta; \delta + 1; z) = \sum_{n=0}^{\infty} \frac{\delta}{\delta+n} z^n. \quad (87)$$

Comparing with (82) and using $\alpha/(\beta\delta) = b_1/\beta = x_+ - x_-$,

$$\Psi(\zeta) = x_+ + (x_+ - x_-) [{}_2F_1(1, \delta; \delta + 1; \zeta) - 1]. \quad (88)$$

(84) together with (88) is the closed form. It is not pretty, but it is a genuine closed form in terms of a standard special function, and the elementary series (82) is what one would actually evaluate. An alternative derivation of the same object, via the integral $\int e^{h\tau}/(p + qe^{k\tau}) d\tau$, is given in [appendix A](#); that route is longer but it produces a general identity of independent use.

Remark 6.2 (Domain of validity). The series (82) converges for $|\zeta| < 1$, and the arguments appearing in (84) are W and $We^{b_1 t}$. In the subcritical intracellular regime $\mu > \beta$ we have $b_1 < 0$ and $x_- = \mu/\beta > 1$, so for $x \in [0, 1]$ the argument W lies in $[0, \beta/\mu] \subset [0, 1)$ and $We^{b_1 t}$ is smaller still: the series representation is valid on the whole of the unit square, which is the case of most interest here. When $\beta > \mu$ the series converges only for x above the smaller fixed point μ/β , and elsewhere one must use the analytic continuation of ${}_2F_1$ — conveniently, the Euler integral

$${}_2F_1(1, \delta; \delta + 1; \zeta) = \delta \int_0^1 \frac{u^{\delta-1}}{1 - \zeta u} du \quad (\delta > 0, \zeta \notin [1, \infty)),$$

which was used for the numerical verification quoted above.

Remark 6.3 (Resonance). The denominators in (82) vanish when $n + \delta = 0$ for some integer $n \geq 1$, that is when

$$\alpha = n(\mu - \beta), \quad n = 1, 2, \dots$$

This is the classical resonance condition for linearising the y -equation: the homogeneous decay rate α of the absorption channel coincides with a harmonic of the intracellular relaxation rate $\mu - \beta$. The generating function itself is perfectly regular there — it is analytic in all parameters — so the singularity is removable, and at resonance the corresponding term acquires a factor τ (a secular term) in place of the simple exponential. In practice the resonant parameter values form a measure-zero set and the limit may be taken numerically, but the degeneracy should be kept in mind when α happens to sit near an integer multiple of $\mu - \beta$.

Appendices

A An integral identity for the hypergeometric function

The route to (82) taken in [section 6.4](#) expands the characteristic $x(\tau)$ in a geometric series and integrates term by term. An alternative is to integrate the quotient (79) directly, splitting the numerator into two pieces, each of the form

$$I(\tau) = \int \frac{e^{h\tau}}{p + qe^{k\tau}} d\tau, \quad (89)$$

with h, k, p, q constant. That route is longer and easier to get wrong — the two pieces carry different prefactors, and dropping one produces a formula that still satisfies the initial condition while violating $G(1, 1, t) = 1$ — but the identity it rests on is useful in its own right, so we record and prove it here.

Proposition .4. For $h/k > 0$ and $-\frac{q}{p}e^{k\tau} \notin [1, \infty)$,

$$\int \frac{e^{h\tau}}{p + qe^{k\tau}} d\tau = \frac{e^{h\tau}}{ph} {}_2F_1\left(1, \frac{h}{k}; \frac{h}{k} + 1; -\frac{q}{p}e^{k\tau}\right) + \text{const.} \quad (90)$$

Proof. We use the Euler integral representation of the hypergeometric function,

$${}_2F_1(a, b; c; z) = \frac{\Gamma(c)}{\Gamma(b)\Gamma(c-b)} \int_0^1 u^{b-1}(1-u)^{c-b-1}(1-zu)^{-a} du, \quad (91)$$

Γ being the standard gamma function.

Start from (89) and substitute $v = e^{k\tau}$, so that $d\tau = k^{-1}v^{-1}dv$:

$$I = \frac{1}{k} \int \frac{v^{h/k}}{p + qv} v^{-1} dv = \frac{1}{pk} \int \frac{v^{h/k-1}}{1 + \frac{q}{p}v} dv.$$

Fix the antiderivative by writing this as a definite integral from 0 to v in a dummy variable w (the integral converges at the lower limit precisely because $h/k > 0$):

$$I = \frac{1}{pk} \int_0^v \frac{w^{h/k-1}}{1 + \frac{q}{p}w} dw.$$

Now rescale, $w = vu$, so that $dw = v du$ with v held fixed:

$$I = \frac{1}{pk} \int_0^1 \frac{(vu)^{h/k-1}}{1 + \frac{q}{p}vu} v du = \frac{v^{h/k}}{pk} \int_0^1 \frac{u^{h/k-1}}{1 + \frac{q}{p}vu} du. \quad (92)$$

The remaining integrand matches that of (91),

$$\int_0^1 u^{h/k-1}(1-u)^0 \left(1 - \left(-\frac{q}{p}v\right)u\right)^{-1} du = \int_0^1 u^{b-1}(1-u)^{c-b-1}(1-zu)^{-a} du,$$

with $b = h/k$, $c = h/k + 1$, $a = 1$ and $z = -\frac{q}{p}v$. Hence

$$\int_0^1 \frac{u^{h/k-1}}{1 + \frac{q}{p}vu} du = \frac{\Gamma\left(\frac{h}{k}\right)\Gamma(1)}{\Gamma\left(\frac{h}{k} + 1\right)} {}_2F_1\left(1, \frac{h}{k}; \frac{h}{k} + 1; -\frac{q}{p}v\right) = \frac{k}{h} {}_2F_1\left(1, \frac{h}{k}; \frac{h}{k} + 1; -\frac{q}{p}v\right),$$

using $\Gamma(z+1) = z\Gamma(z)$. Substituting into (92),

$$I = \frac{v^{h/k}}{ph} {}_2F_1\left(1, \frac{h}{k}; \frac{h}{k} + 1; -\frac{q}{p}v\right),$$

and restoring $v = e^{k\tau}$ gives (90). $\square$

Remark .5. Applied to the two pieces of (79) — the first with $h = \alpha$, the second with $h = \alpha + b_1$, and both with $k = b_1$, $p = (s - x_-)$, $q = -(s - x_+)$ — [proposition .4](#) reproduces (88) after the prefactors $\alpha x_+(s - x_-)$ and $-\alpha x_-(s - x_+)$ are carried through correctly. The factor $(s - x_+)/(s - x_-)$ attached to the second piece is easily lost, and its loss is invisible at $t = 0$; checking $G(1, 1, t) = 1$ catches it immediately.

B Extracting state probabilities from a generating function

A generating function defined by a formula rather than as an explicit series — for example

$$G(t; x, y) = \left((1 - e^{-\alpha t})x + e^{-\alpha t}y \right)^n \quad (93)$$

— can always be converted back to the form

$$G(t; x, y) = \sum_{i=0}^{\infty} \sum_{j=0}^{\infty} p_{i,j}(t) x^i y^j, \quad (94)$$

recovering the time-dependent state probabilities. All that is required is that G be repeatedly partially differentiable in x and y .

The reason is the bivariate Maclaurin expansion,

$$f(x, y) = \sum_{k=0}^{\infty} \frac{1}{k!} \sum_{\rho=0}^k \binom{k}{\rho} \partial_x^\rho \partial_y^{k-\rho} f \Big|_{(0,0)} x^\rho y^{k-\rho},$$

whose coefficient of $x^i y^j$ (taking $k = i + j$, $\rho = i$) is $\frac{1}{(i+j)!} \binom{i+j}{i} \partial_x^i \partial_y^j f \Big|_{(0,0)}$. Hence

$$p_{i,j}(t) = \frac{1}{i! j!} \partial_x^i \partial_y^j G \Big|_{(0,0)}. \quad (95)$$

The multilinear case

Both (55) and (59) have the form

$$G(t; x, y) = (c(t) + f(t)x + g(t)y)^n, \quad (96)$$

for which the derivatives are immediate. Each differentiation lowers the power by one and brings down a factor, so

$$\partial_x^a \partial_y^b G = \frac{n!}{(n-a-b)!} f^a g^b (c + fx + gy)^{n-a-b} \quad (a + b \leq n), \quad (97)$$

and zero for $a + b > n$. Evaluating at the origin and applying (95),

$$p_{i,j}(t) = \frac{n!}{i! j! (n-i-j)!} f(t)^i g(t)^j c(t)^{n-i-j}, \quad i + j \leq n, \quad (98)$$

and $p_{i,j} = 0$ otherwise. This is the trinomial distribution, which is exactly what it should be: the n particles are independent, and at time t each is in one of three states with probabilities c (neither inside nor outside — that is, dead), f (inside) and g (outside). Note in particular that the falling factorial $n!/(n-i-j)!$ in (97) is *not* the same as n^{i+j} ; the two agree only to leading order in n , and using the latter breaks normalisation.

Applied to the absorption models

For absorption only, (93), we have $c = 0$, $f = 1 - e^{-\alpha t}$ and $g = e^{-\alpha t}$. The factor c^{n-i-j} then vanishes unless $i + j = n$, and (98) gives

$$p_{i,j}(t) = \binom{n}{i} (1 - e^{-\alpha t})^i (e^{-\alpha t})^j, \quad i + j = n,$$

so $\mathbf{X}_t \sim \text{Binomial}(n, 1 - e^{-\alpha t})$ and $\mathbf{Y}_t = n - \mathbf{X}_t$, as anticipated in [section 6.1](#).

For absorption–death, [\(59\)](#), the three functions are

$$c(t) = 1 - \frac{\mu}{\mu - \alpha} e^{-\alpha t} + \frac{\alpha}{\mu - \alpha} e^{-\mu t}, \quad f(t) = \frac{\alpha}{\mu - \alpha} (e^{-\alpha t} - e^{-\mu t}), \quad g(t) = e^{-\alpha t},$$

which are precisely $p_{0,0}$, $p_{1,0}$ and $p_{0,1}$ of [\(58\)](#), and

$$p_{i,j}(t) = \frac{n!}{i! j! (n - i - j)!} f(t)^i g(t)^j c(t)^{n-i-j}.$$

Each particle independently ends up dead, absorbed, or still outside, with those three probabilities; the joint law is trinomial. That the generating function factorises in this way is of course just [\(54\)](#) read backwards.

The absorption–birth–death generating function of [proposition 6.1](#) is not of the form [\(96\)](#), and there [\(95\)](#) must be applied to [\(84\)](#) directly, or the coefficients extracted numerically by Cauchy integrals on circles $|x| = |y| = \rho < 1$.

References

- [1] Krishna B. Athreya and Peter E. Ney. *Branching Processes*. Springer, Berlin, 1972.
- [2] Theodore E. Harris. *The Theory of Branching Processes*. Springer, Berlin, 1963.
- [3] Akiva M. Yaglom. Certain limit theorems of the theory of branching random processes. *Doklady Akademii Nauk SSSR (N.S.)*, 56:795–798, 1947.
- [4] Graham E. Budd and Richard P. Mann. History is written by the victors: The effect of the push of the past on the fossil record. *Evolution*, 72(11):2276–2291, 2018.
- [5] Linda J. S. Allen. *An Introduction to Stochastic Processes with Applications to Biology*. CRC Press, Boca Raton, FL, 2nd edition, 2010.
- [6] Kenneth O Stanley and Risto Miikkulainen. Evolving neural networks through augmenting topologies. *Evolutionary Computation*, 10(2):99–127, 2002.
- [7] Benjamin G Cohen, Burcu Beykal, and George M Bolas. Physics-informed genetic programming for discovery of partial differential equations from scarce and noisy data. *Journal of Computational Physics*, 514:113261, 2024.
- [8] Gabriel Koenigs. Recherches sur les intégrales de certaines équations fonctionnelles. *Annales Scientifiques de l’École Normale Supérieure*, 1:3–41, 1884.
- [9] Paul-Georg Becker and Walter Bergweiler. Hypertranscendence of conjugacies in complex dynamics. *Mathematische Annalen*, 301(1):463–468, 1995.